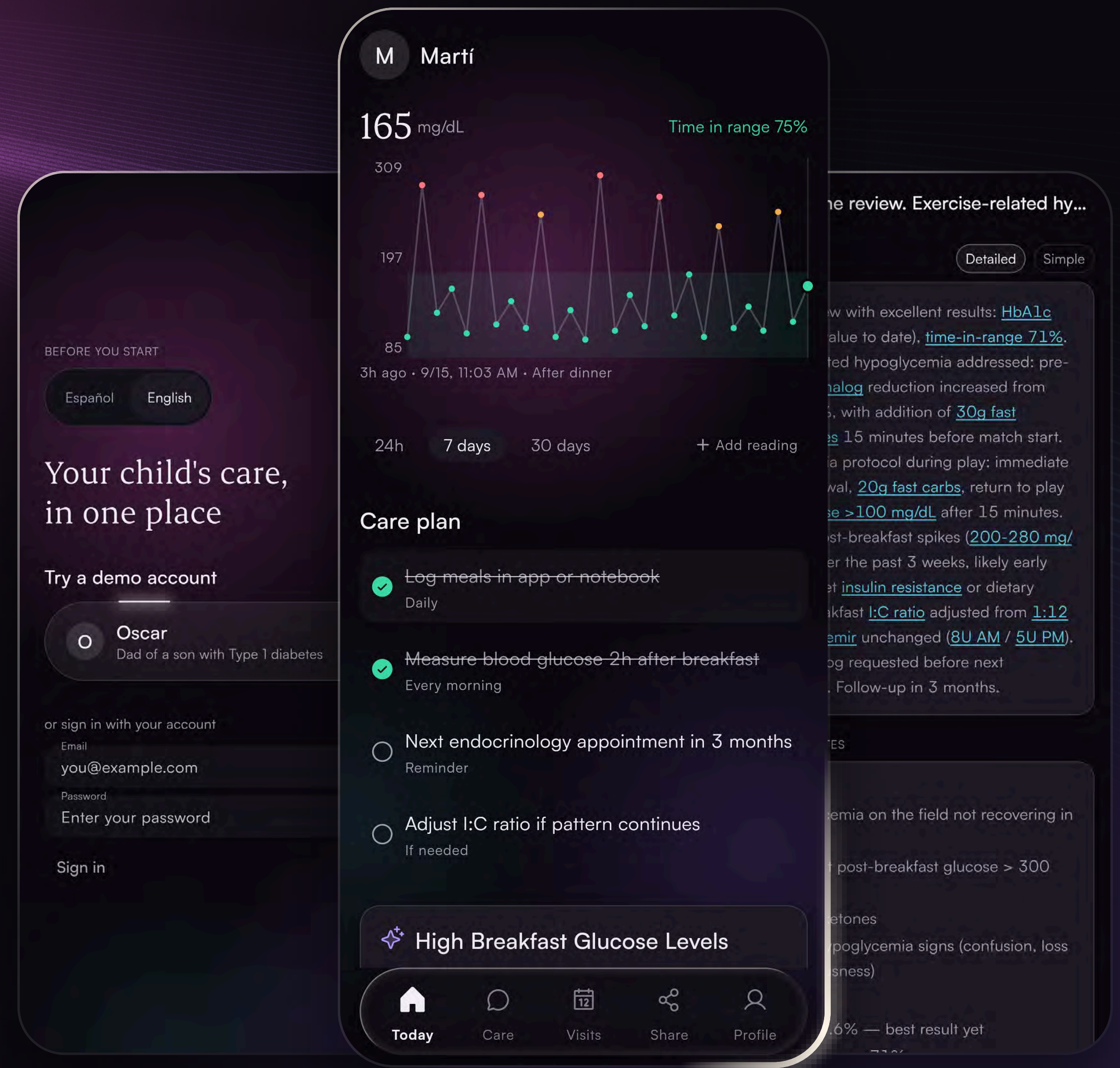


Pocket Pediatrics

Samhita Bommakanti • Carly May Larsen

Francisco Moraschi • Çağdaş Ergenç



Caring for a medically complex child
isn't just hard on the child.

62%

of medically complex children
are readmitted within 90 days of discharge

Readmission rate

49%

of caregivers of children with
chronic disease report a very severe burden

Caregiver burden

249hrs

lost per year to scheduling and admin tasks
Parents spend ~32hrs per week on mental load

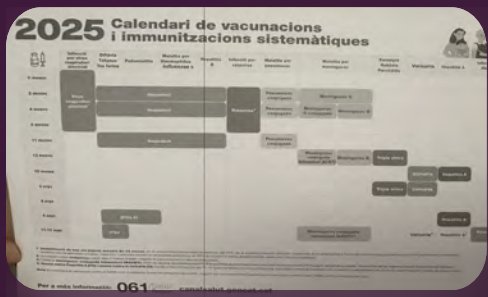
Admin Tasks

Research Process

Six stages, from broad signals to a validated prototype.

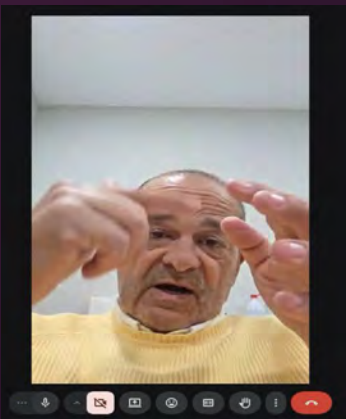
Community Outreach

Surveys & posters



Interviews

Qualitative depth



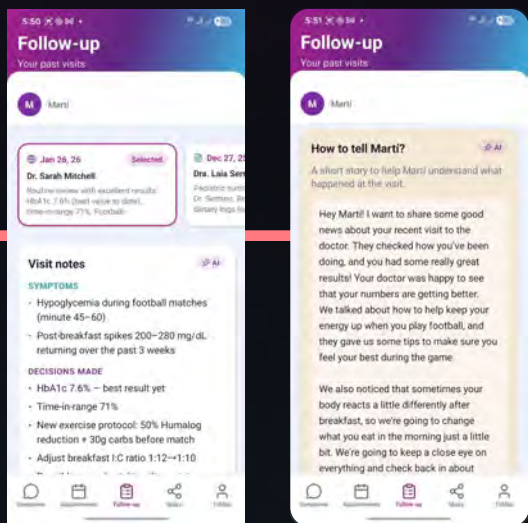
Data Analysis & Journey Mapping

Reveal hidden behaviours



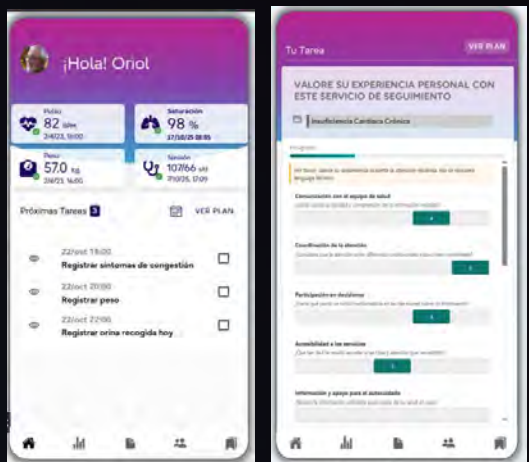
Prototype Iteration

Refine & validate



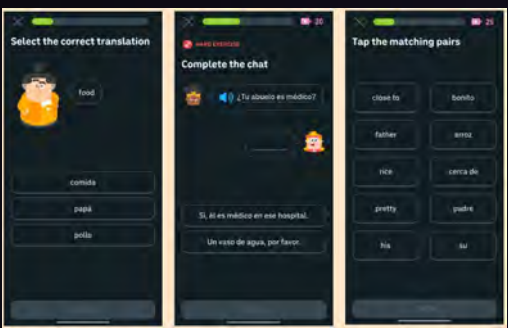
Prototype & User Testing

14 participants, 5 flows



Sacrificial Concepts

Test ideas early



THE LEAD INSIGHT

Cognitive load —
is the core stressor.

This shaped every design decision we made.

Meet the Caregivers



Juggling Jesse

Mom of Two Under Three

30 Y/O

Graphic Designer

Barcelona

HER COGNITIVE LOAD

Health info scattered across apps, chats, paper, and calendars — nothing in one place.

PAIN POINTS

- Remembering what appointments are needed, and when
- Repeating full medical history at every visit
- Sharing the care load with her partner

"Right now I have three different tools to track everything."



Overwhelmed Oscar

Single Dad of Son with Diabetes

35 Y/O

Project Manager

Barcelona

HIS COGNITIVE LOAD

No guidance between appointments - every escalation is a solo decision with no safety net.

PAIN POINTS

- Medical jargon leaves him unsure of next steps
- No way to track whether things are improving
- Just waiting and hoping between visits

"The hardest part is not knowing if it's a bad day — or an emergency."

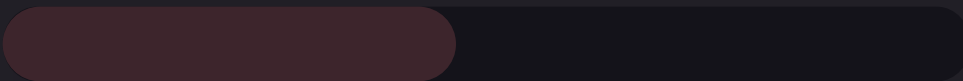
Managing a chronic disease costs caregivers ~32 hours a week of mental load.

The Insights

Six statistics across three insights — all pointing to one overloaded caregiver.

Reduce Cognitive Load

Time spent weekly managing care



40%+

of caregivers of medically complex children

Median doctor visits per year



11–15 typical visit range for medically complex children

Pub Med [National Profile of Caregiver Challenges](#)

Speak Their Language

Appointment information recalled



49%

of information was roughly remembered

Visits with unexplained jargon



80%

of visits leave medical terms unexplained

Parents with low health literacy



1 in 4

rises to 1 in 3 in emergency settings

Source: [Surgeon Use of Medical Jargon with Parents in the Outpatient Setting \(PMC6525640\)](#)

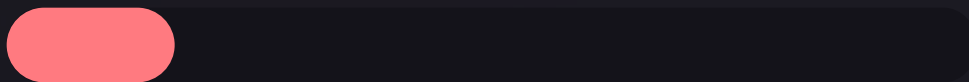
Centralize Information

Parents needing coordination help



67% of parents reported needing care coordination support

Forgone medical care



15.8%

when caregivers spend 5+ hrs/week coordinating

Source: [Inequities in Care Coordination](#)

How Might We

How might we reduce caregivers' cognitive load by redesigning the FHS experience so parents leave each visit with understandable summaries, clear next steps, and a child-friendly story that helps kids understand their diagnosis?

The real gap isn't access to care.

It's what happens between appointments — fragmented info, jargon parents can't act on, children who don't understand.

249 hrs

lost per year
to scheduling & admin tasks

49%

of visit info forgotten
roughly remembered after leaving

67%

need coordination help
of parents, to manage care

Our solution

An AI-powered care assistant embedded within FHS

"What if every caregiver left each appointment with a clear plan and the confidence to act on it?"

01

Appointment Management

Easy to use appointment booking tools that integrate with caregivers' calendars.

02

Follow-Up Instructions

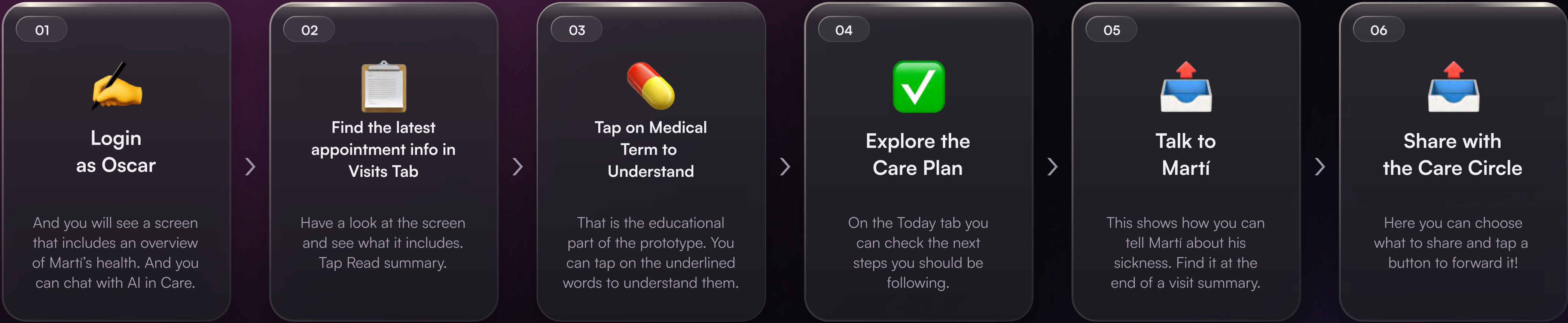
A clear, jargon-free record of every visit. Structured notes the whole care circle can understand and act on.

03

Sharing Tools

A space to share the appointment follow-up and next steps with loved ones. Including a children's story to easily translate for your kids

See it in Action



Working prototype · Flutter + Supabase
Pocket Pediatrics



pocpedv2.netlify.app

SEE IT IN ACTION • 01 / 06

Start on Today

- 1

Log in as Oscar

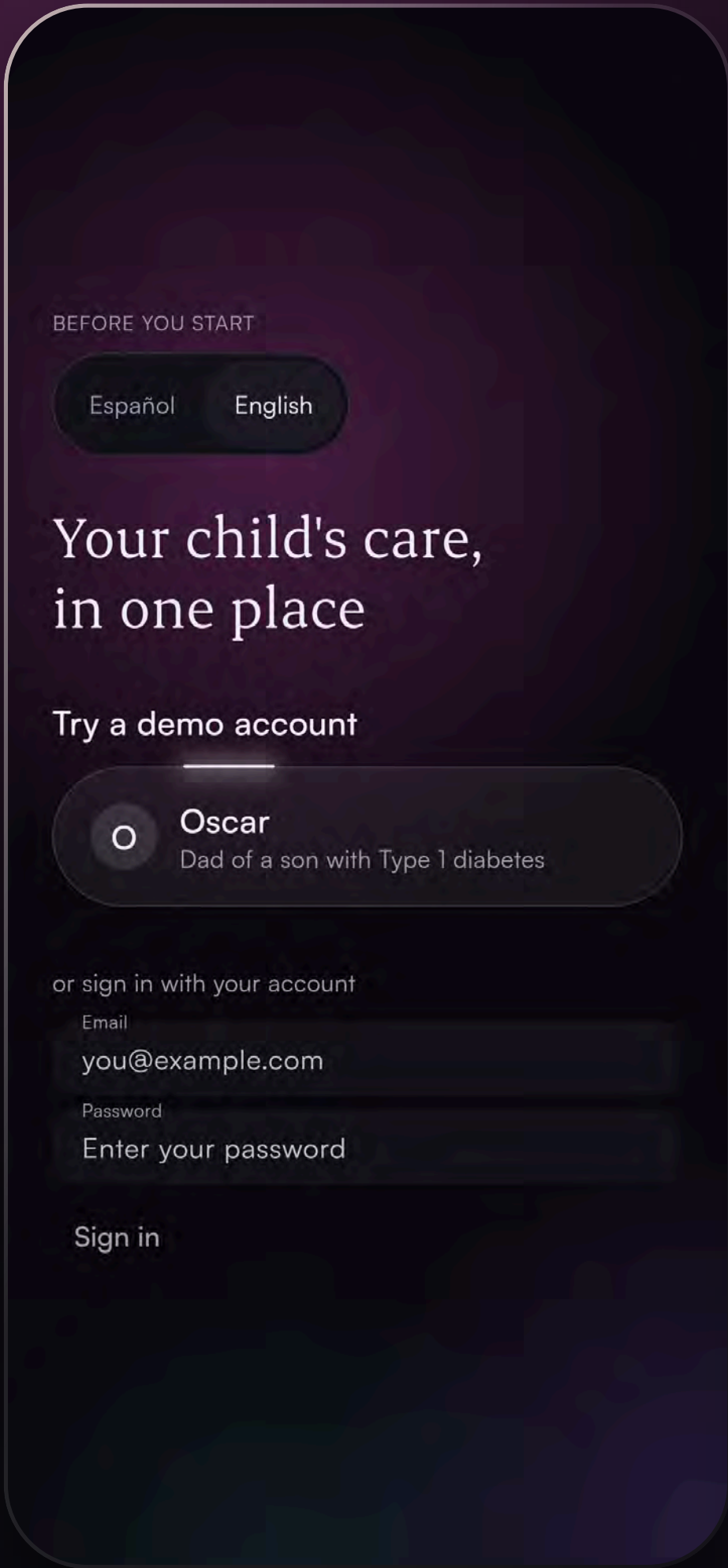
One tap on the demo account, no password.
- 2

See Martí's day at a glance

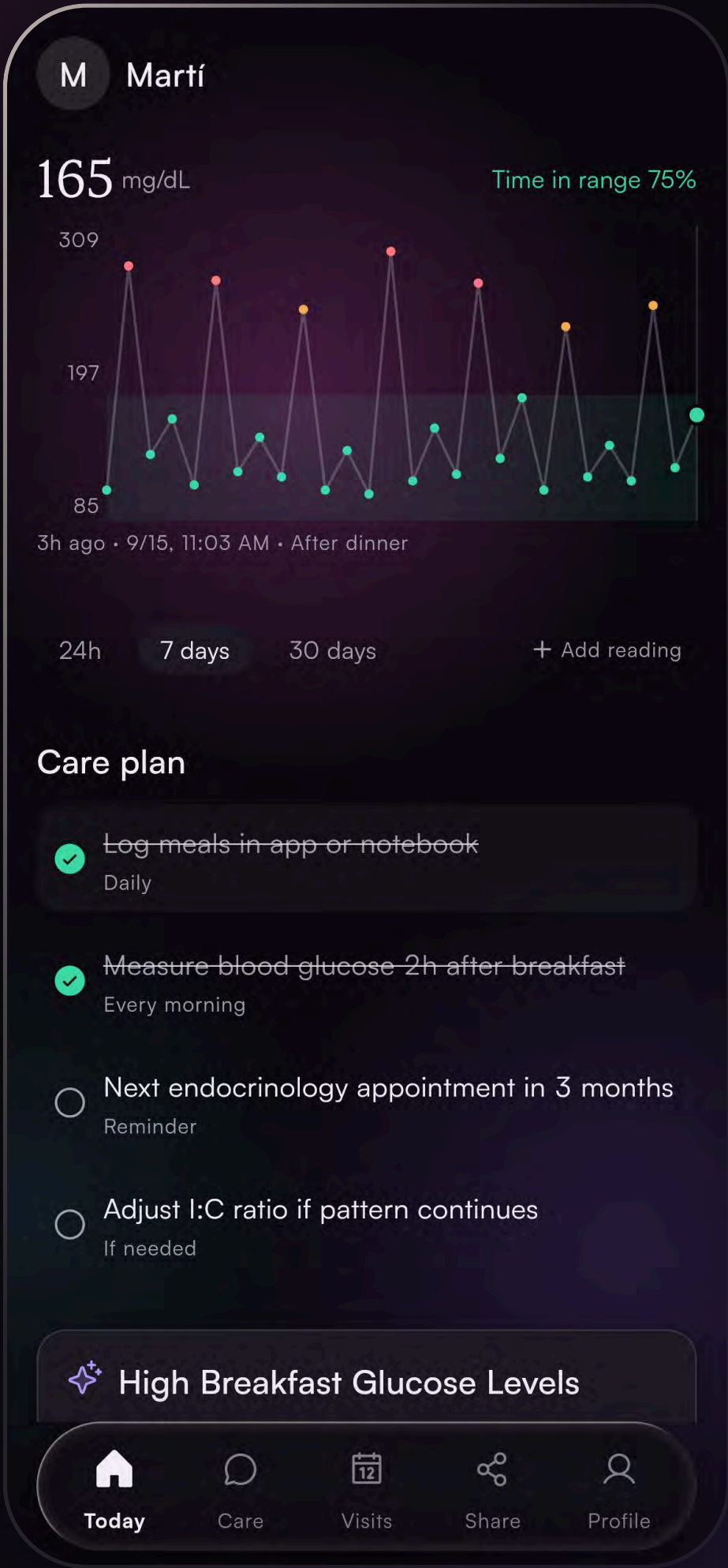
Glucose trend, time in range and the care plan.
- 3

Log a reading

Add reading takes the value, when it was, and a note.



Log in



Today



Add reading

Ask Care before you worry

- 1

Start from a suggestion

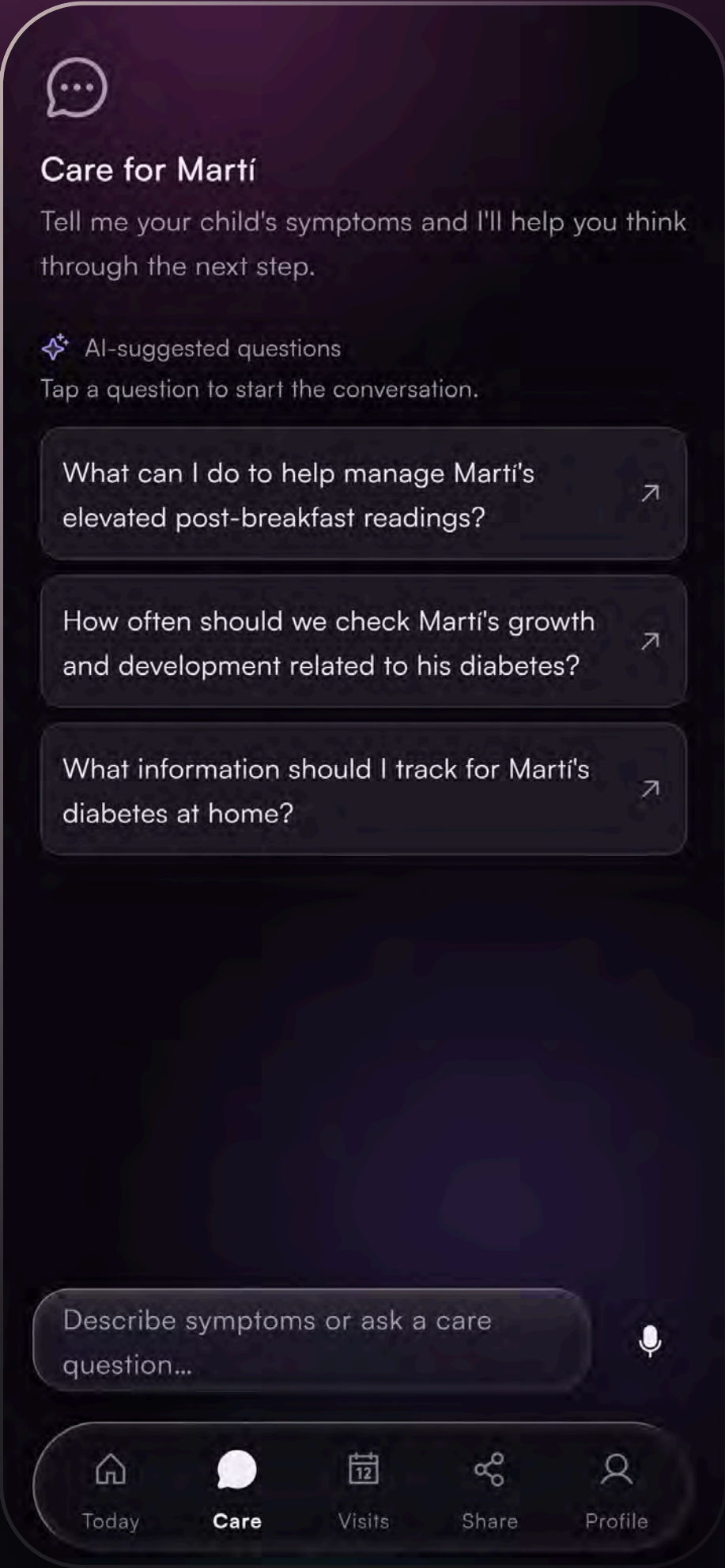
Questions are drafted from Marti’s own readings.
- 2

Describe what you see

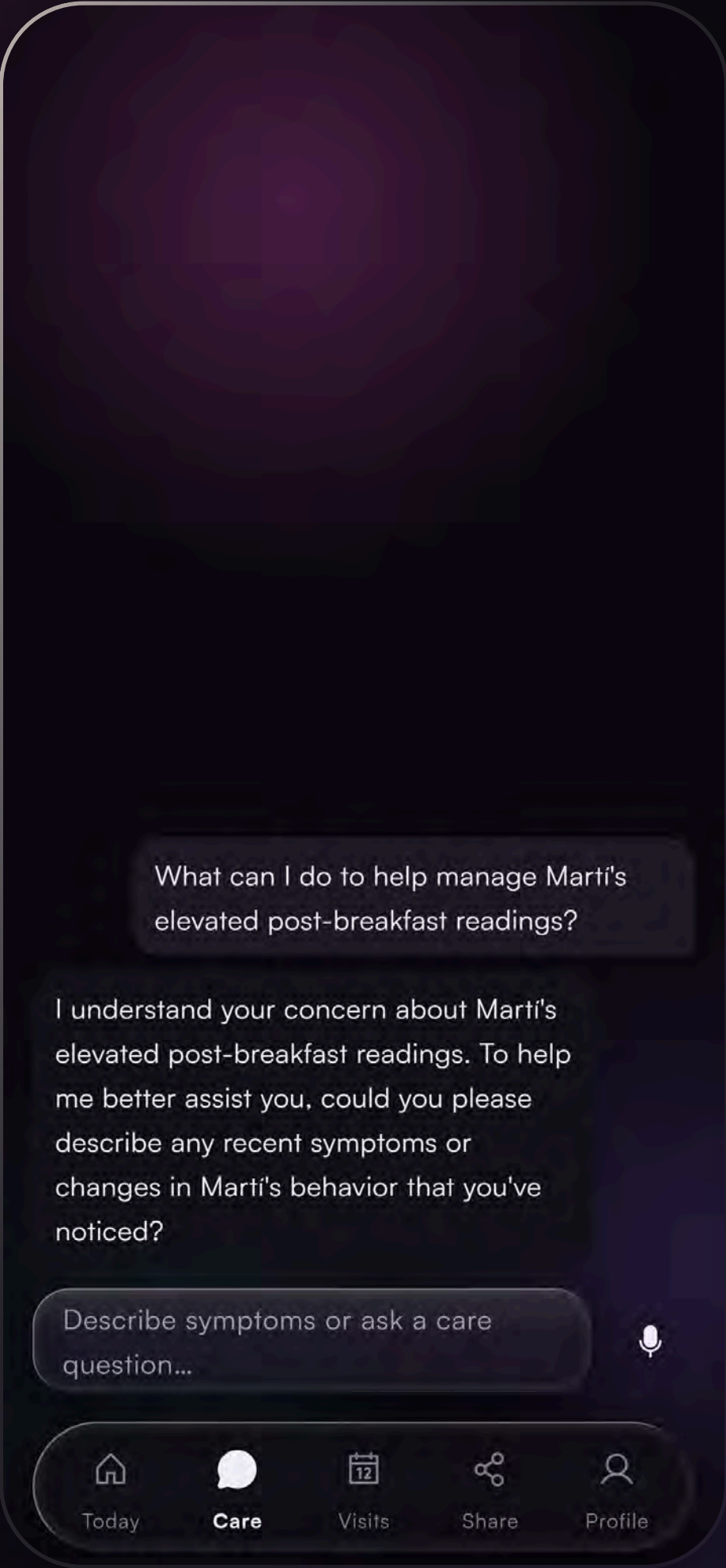
Type or speak. The assistant asks one question at a time.
- 3

Know the next step

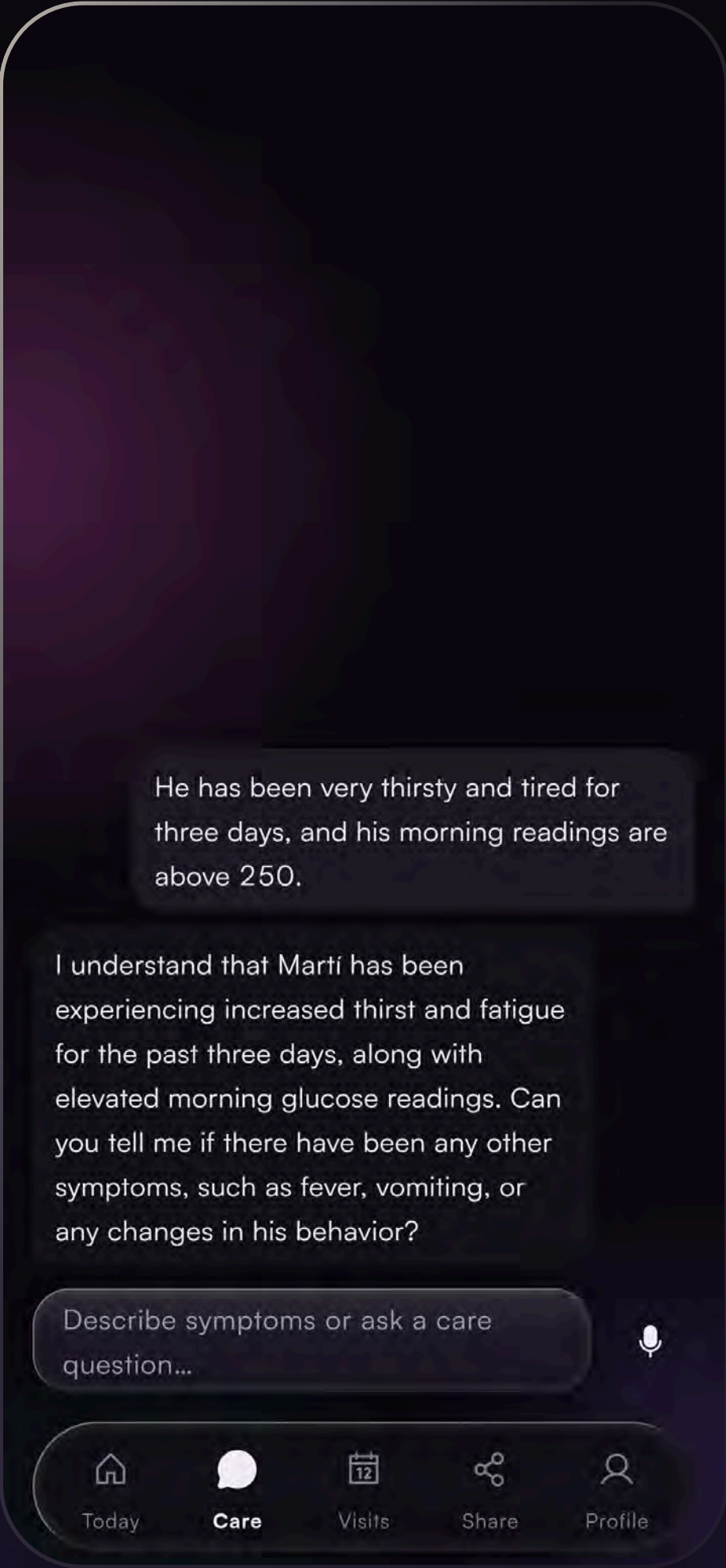
When it has enough, it tells you whether to book a visit.



Suggestions



First question



Follow-up

Understand the visit

- 1

Open the latest visit

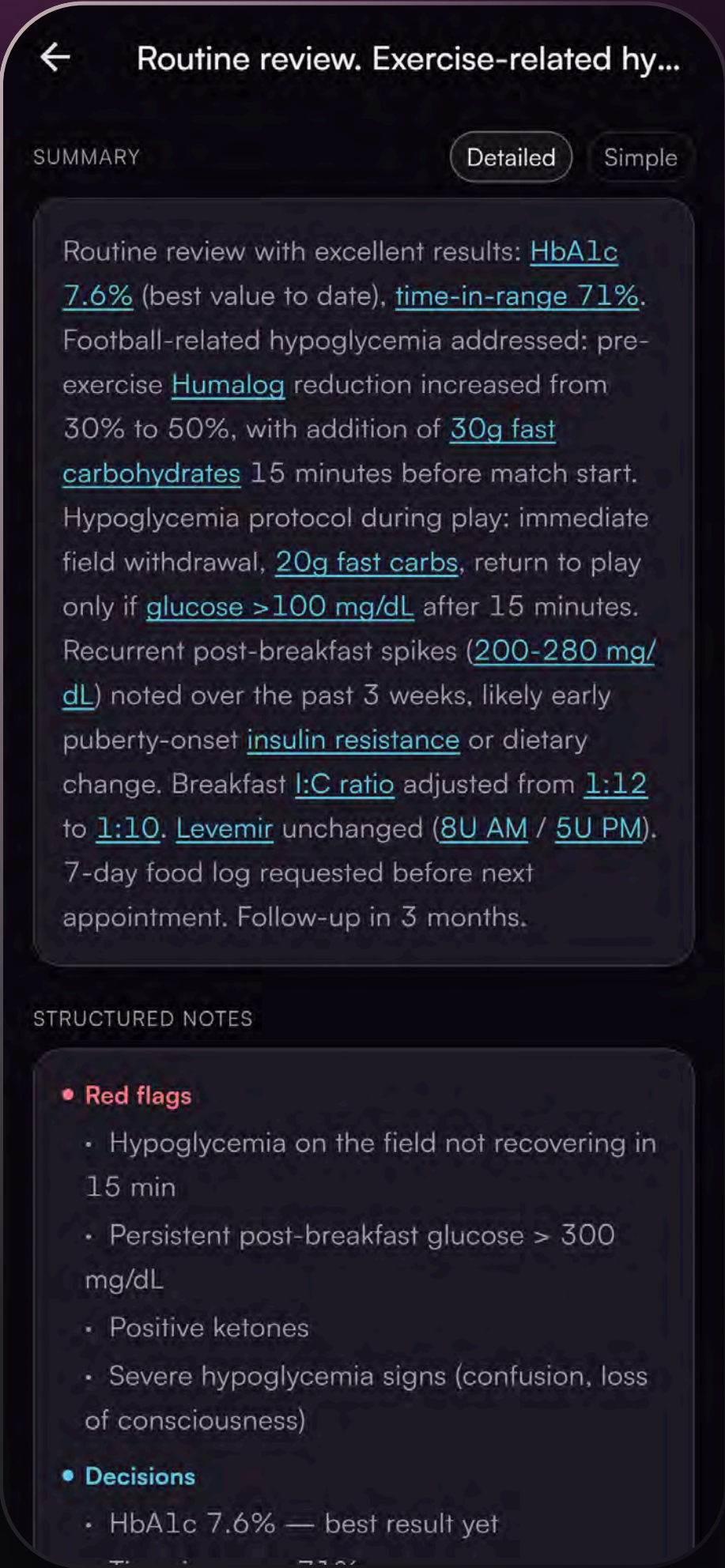
Visits → Read summary.
- 2

Tap a medical term

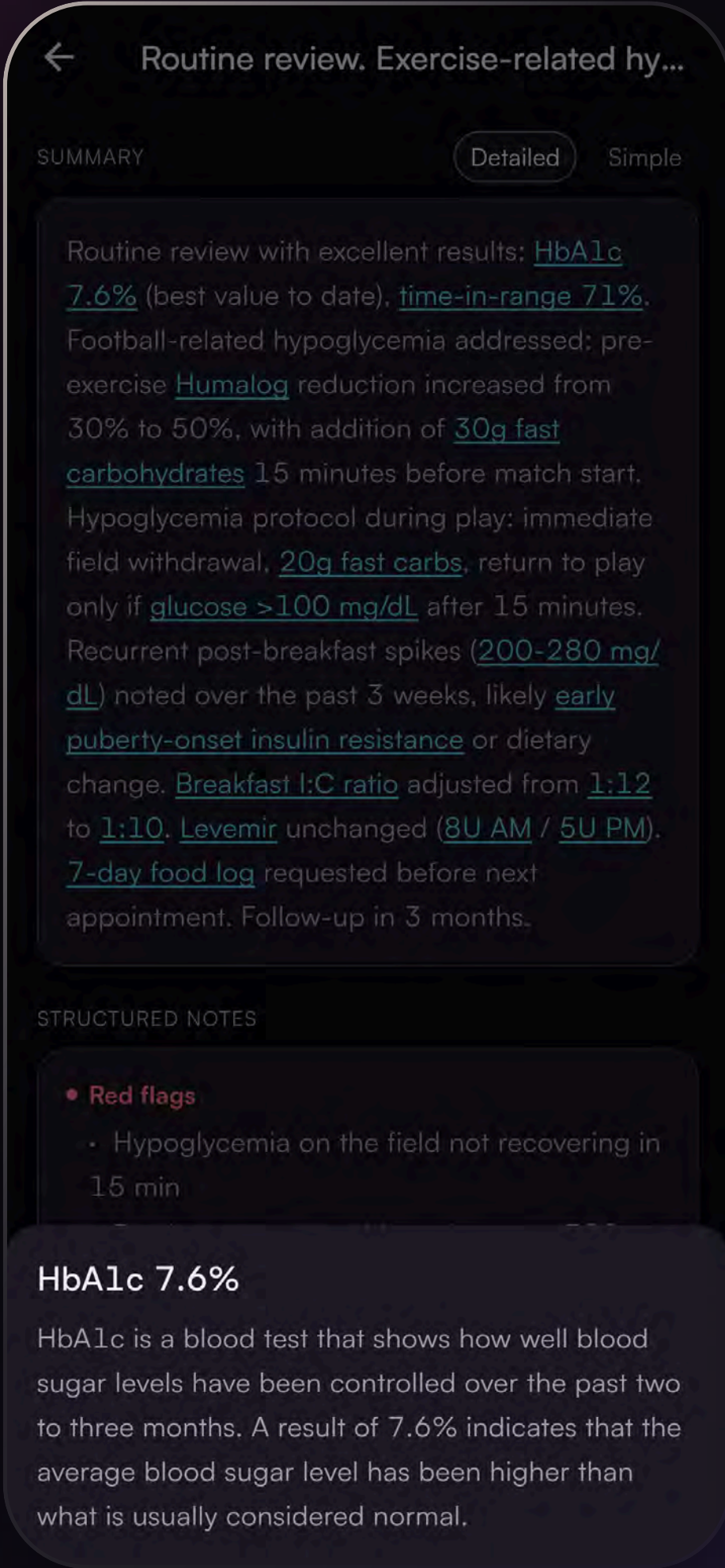
Underlined words open a plain-language explanation.
- 3

Scan the structured notes

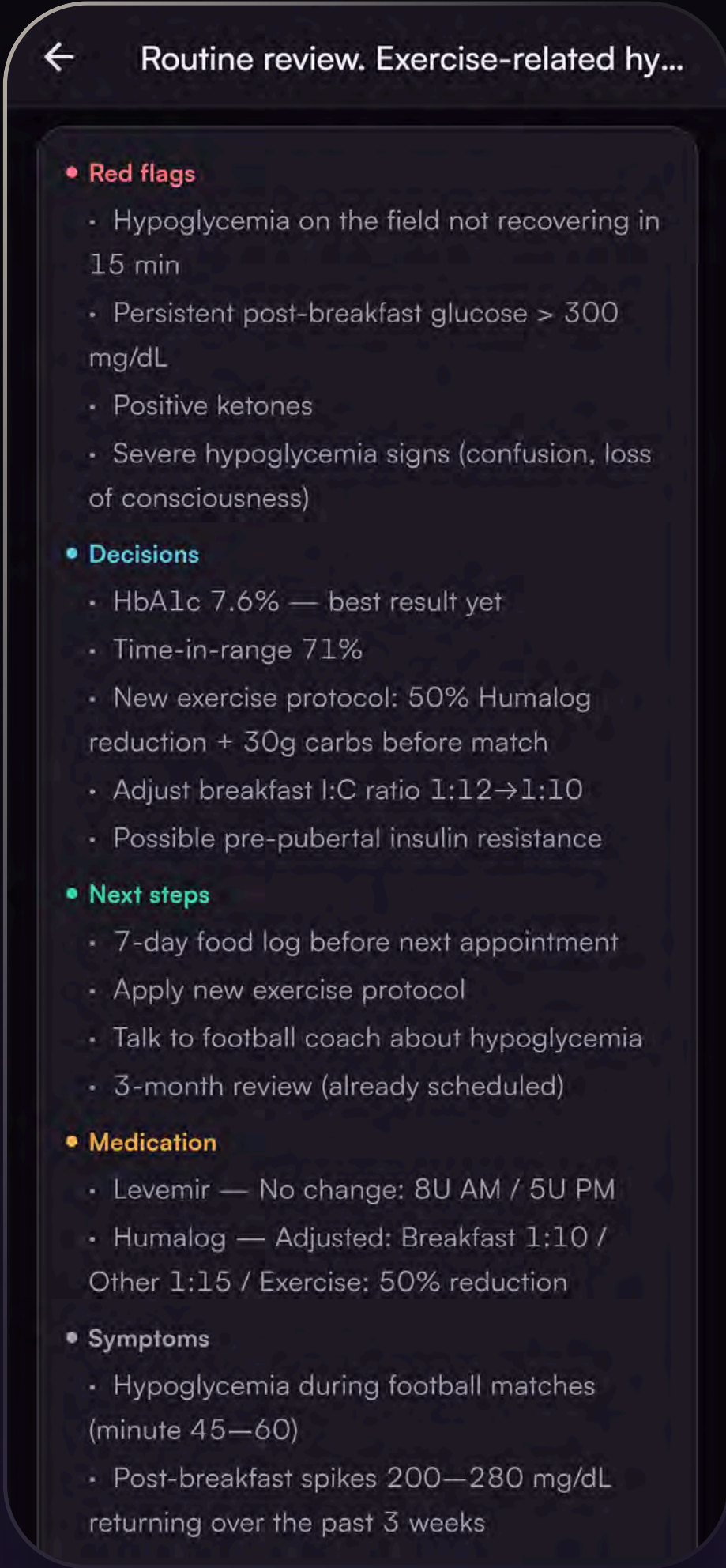
Red flags, decisions, next steps and medication.



Summary



Term explained



Structured notes

SEE IT IN ACTION • 04 / 06

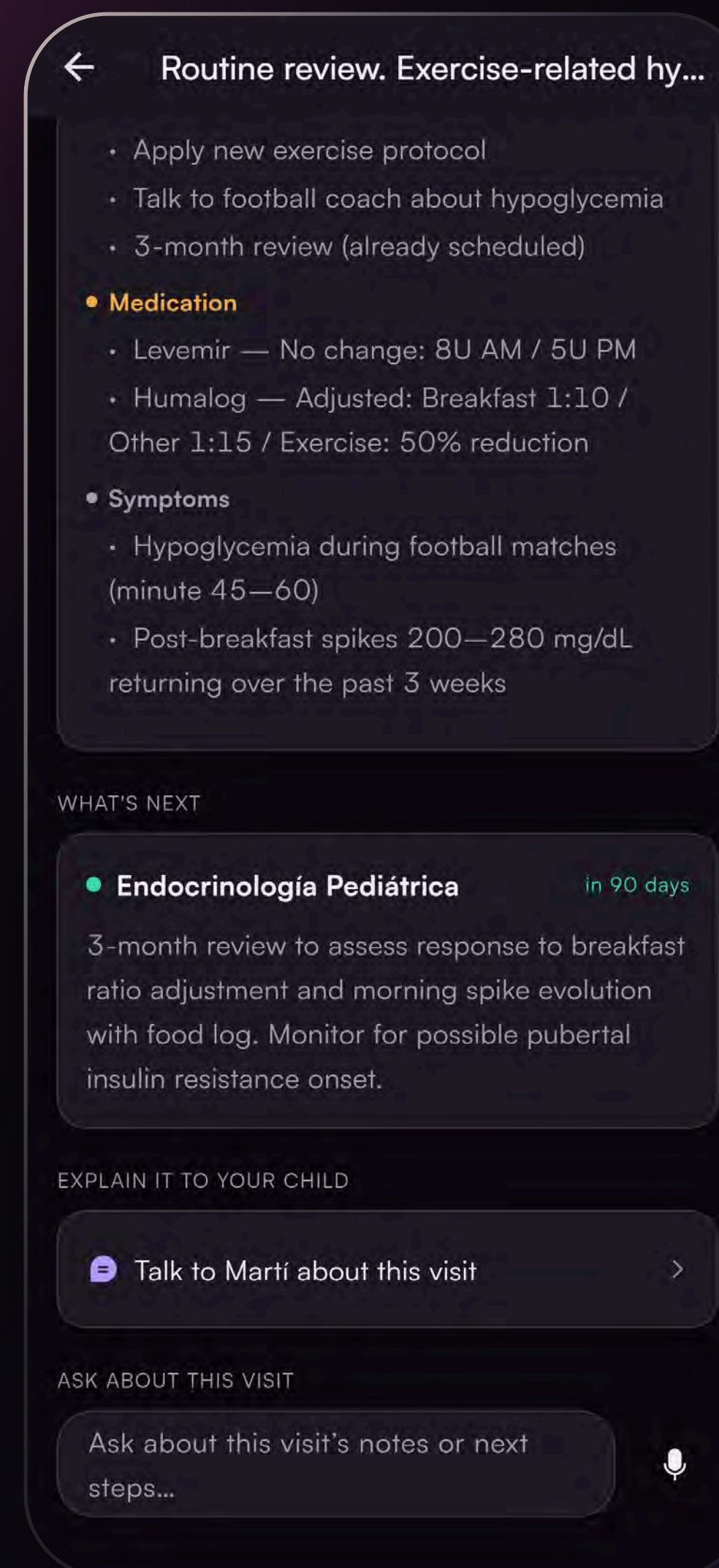
Explain it to Martí

1 See what's next

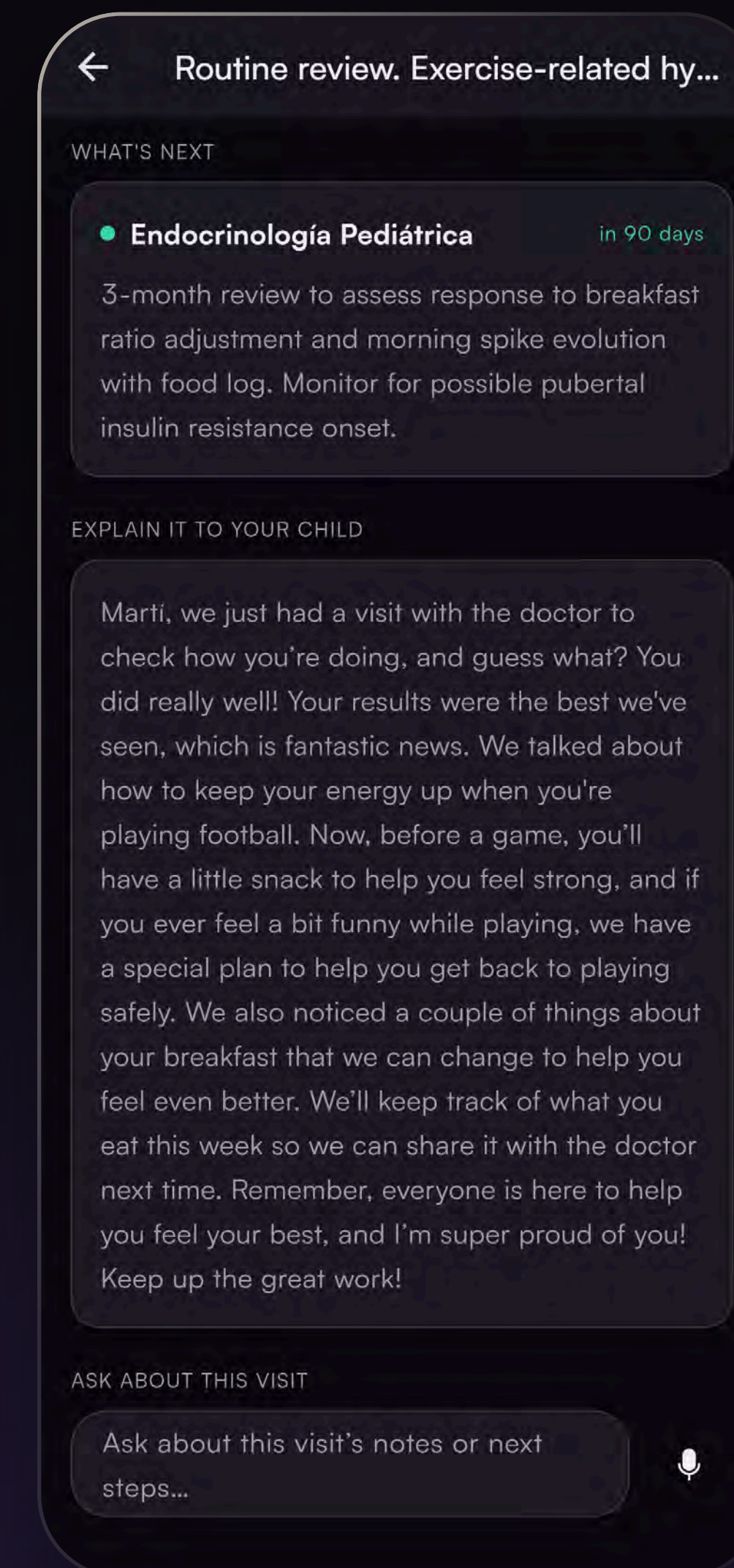
The follow-up appointment and what it will check.

2 Talk to Martí

A short story that explains the visit in words a child understands.



What's next



Talk to Martí

SEE IT IN ACTION • 05 / 06

Share with the care circle

- 1

Choose a visit

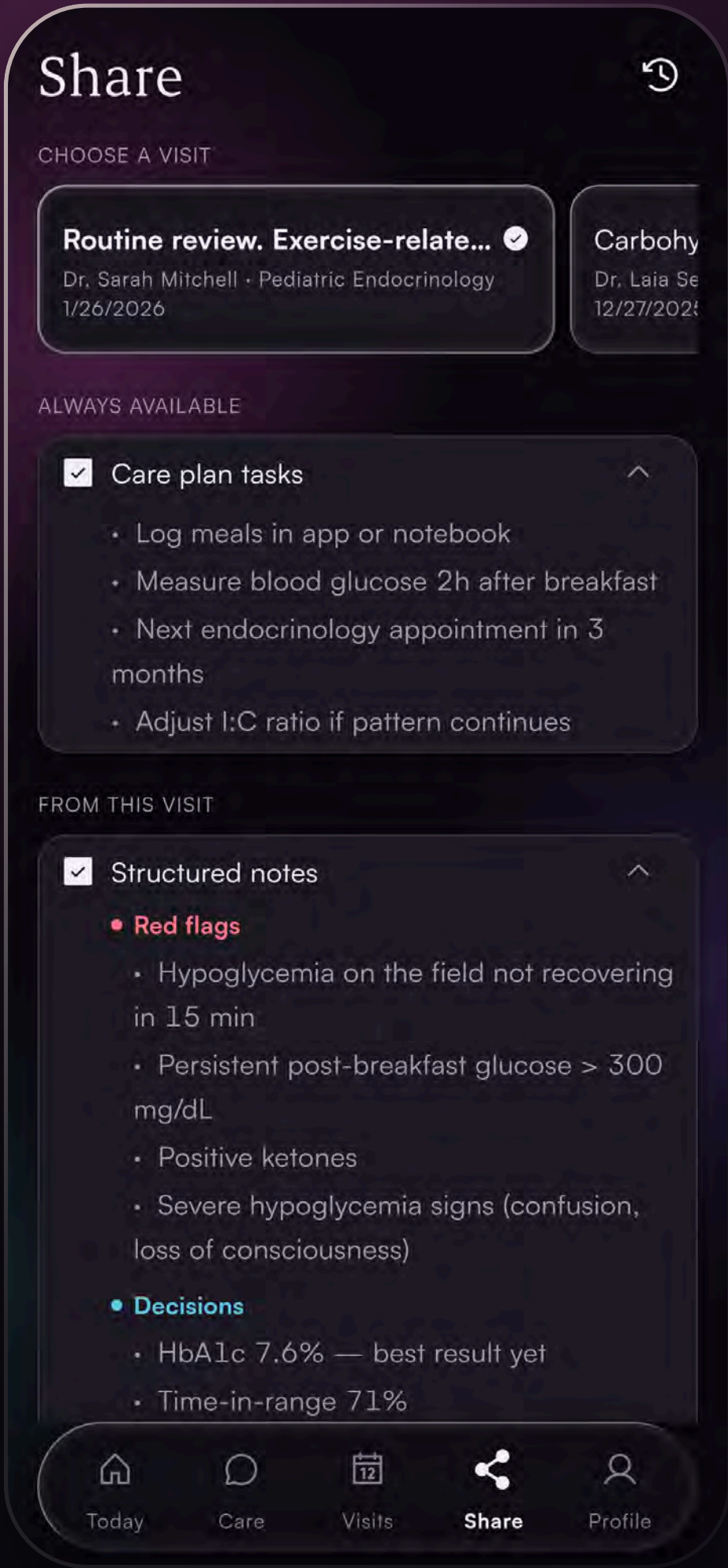
Pick the appointment you want to share.
- 2

Choose what to include

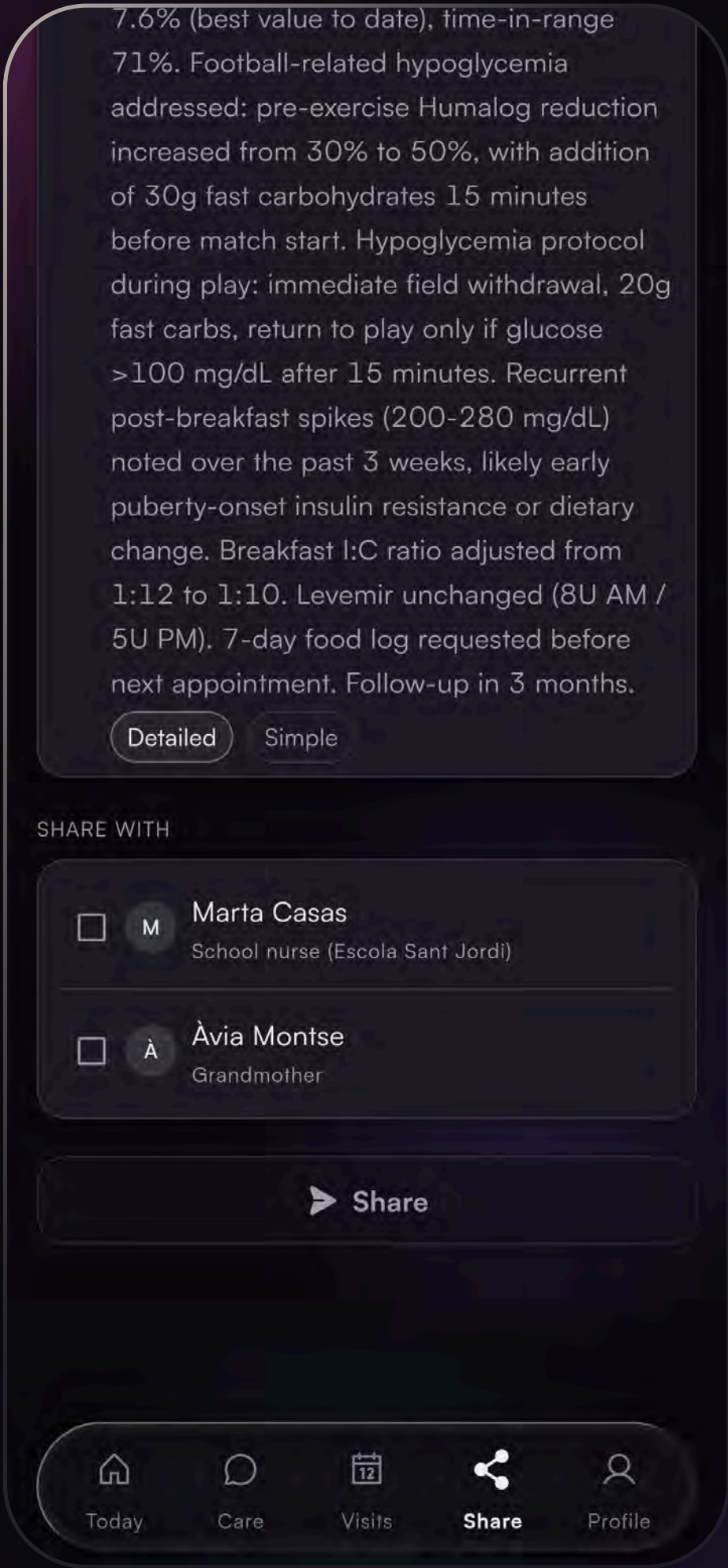
Each section shows its real content before you send.
- 3

Send and keep a record

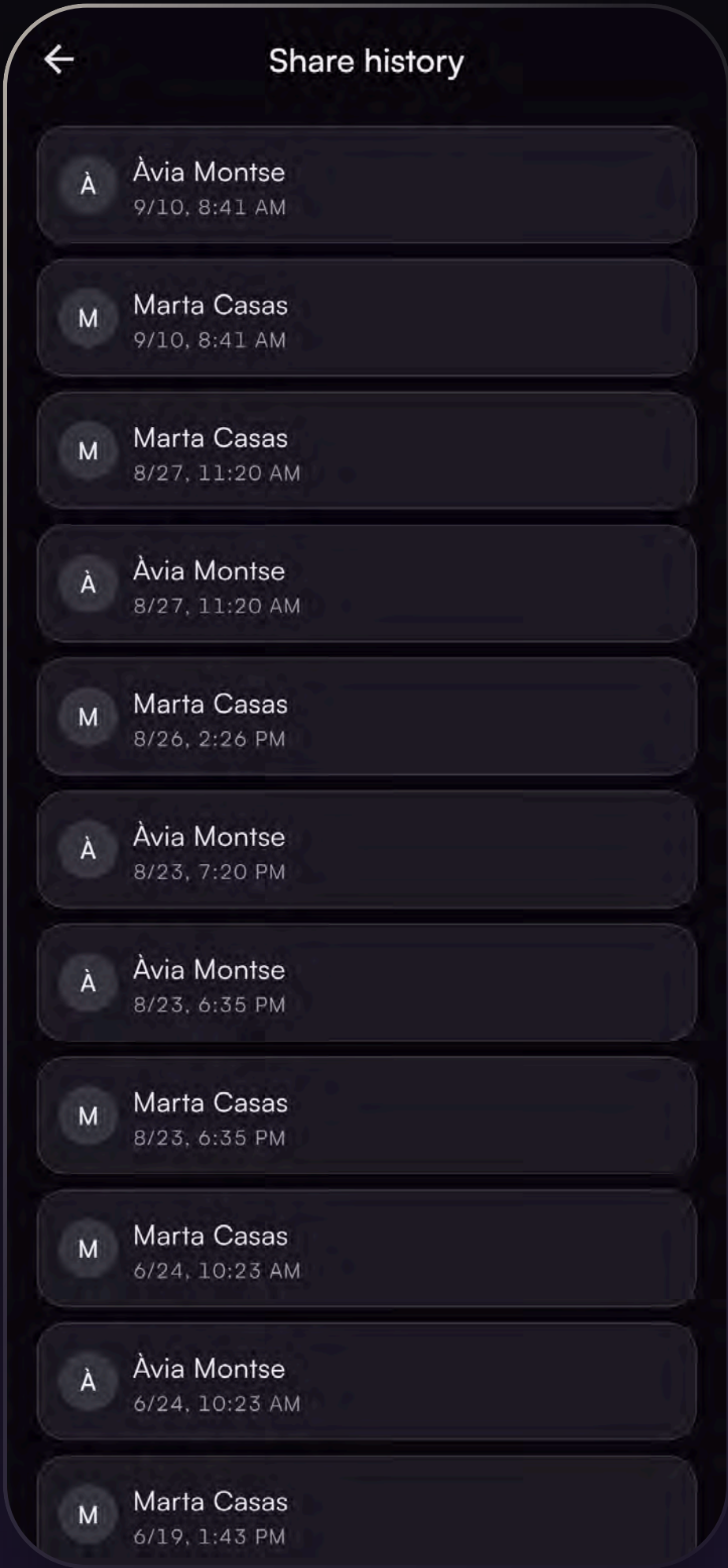
Share history shows who you sent to, and when.



Choose a visit



Share with



Share history

SEE IT IN ACTION • 06 / 06

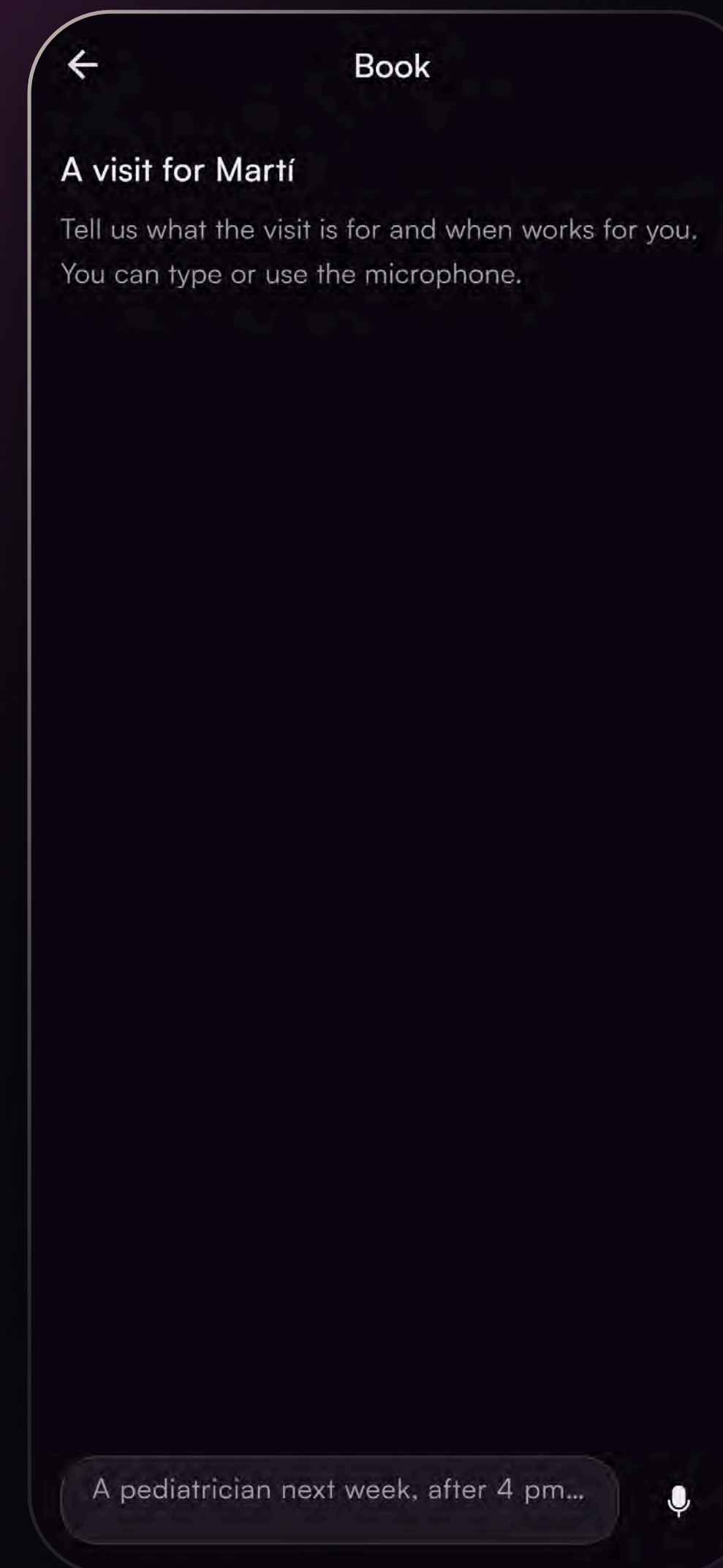
Book and record the next visit

1 Book in plain words

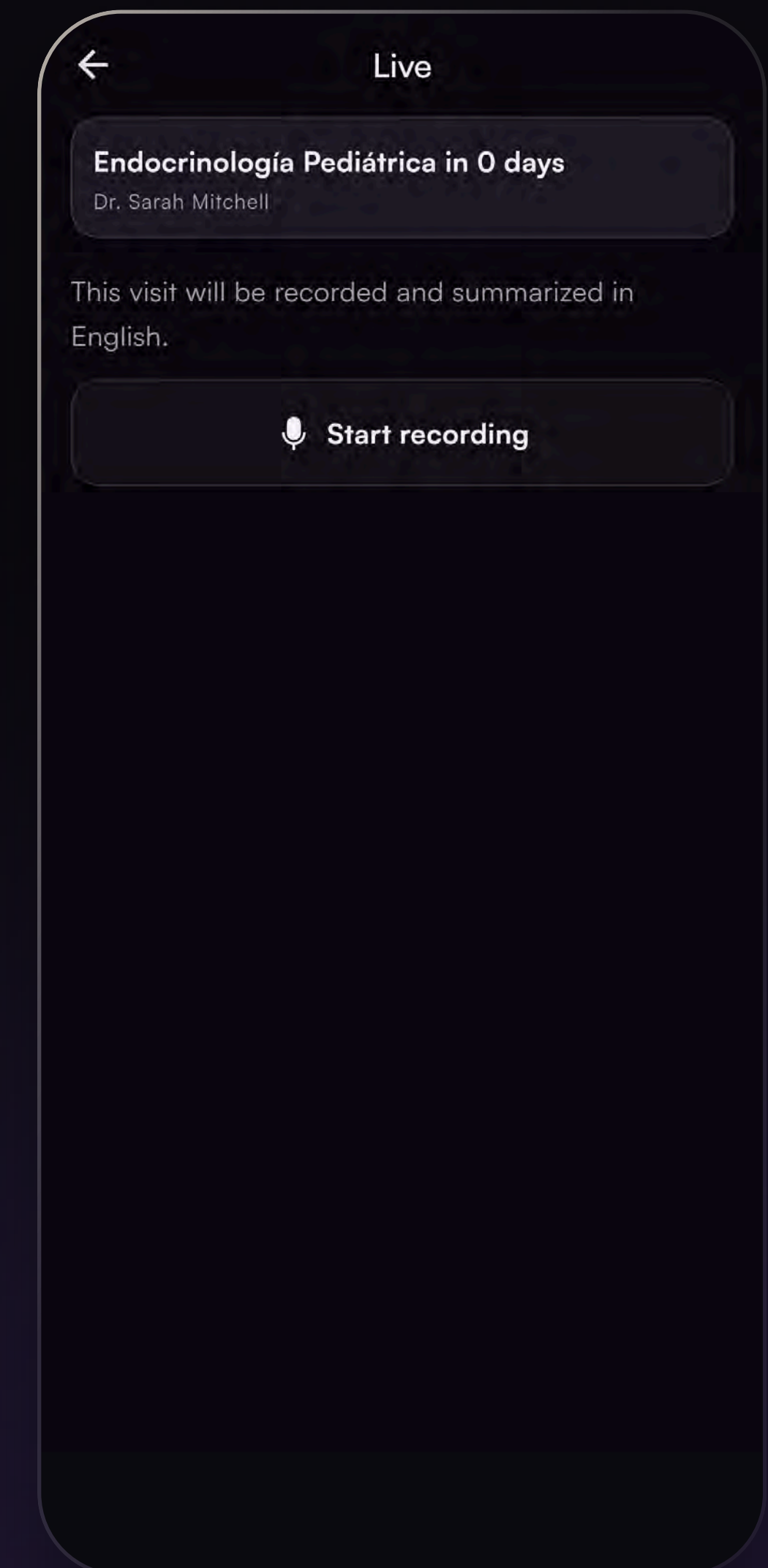
Type or say what you need, like “a pediatrician next week, after 4 pm”.

2 Record the visit

Start the visit to record it and get the summary in your language.



Book



Live visit

Tested & Validated

Three additions that each close a cognitive load gap in FHS today.

FHS TODAY	THE GAP	POCKET PEDIATRICS
APPOINTMENT MANAGEMENT		
Appointment assistance	No caregiver view	Smart scheduling & management ↓ removes the 249 hrs/yr admin burden
RESULTS & FOLLOW-UP		
Chronic follow-up	Medical language barrier	Translated summaries & kid-friendly stories ↓ closes the 49% recall gap
SHARING WITH YOUR CARE CIRCLE		
Detailed care plans	No way to share with your care circle	Sharing tools & custom profiles ↓ solves the 67% coordination need

All three validated, 14 caregivers, 5 user flows.

Every feature scored above 4.0 / 5, and every one reduces caregiver cognitive load.

Every feature reduces one thing: cognitive load.

Our first demo proved the concept. These KPIs are what we'll measure next.

The first demo validated usability and feature concept not yet time saved or comprehension. Below is the direction we are committed to measuring.

Time Saving & Workflow Efficiency

INITIAL RESULT (DEMO)

4.8 / 5

Sharing feature rated #1, all 14 scored 4 or 5

FUTURE TARGET

↓ 249 hrs

per year, in scheduling & admin time per caregiver

Measured via pre/post time-tracking

Caregiver Understanding & Satisfaction

INITIAL RESULT (DEMO)

4.6 / 5

Summary clarity & completeness, above avg

FUTURE TARGET

↑ 80%+

comprehension score after AI translation rollout

Measured via pre/post comprehension quiz

Care Plan Adherence & Follow-Through

INITIAL RESULT (DEMO)

4.0 / 5

Story feature, universally loved

FUTURE TARGET

↑ Adherence

improvement in follow-through, medication & next steps

Measured via app engagement + outcomes

Future Actions & Recommendations

● Already Fixed

Before next test round

- 1 Rename Story → 'Talk to [Child's Name]'
- 2 Fix graph: tooltips, color logic, daily zoom
- 3 Rename Appointments tab to "Appointments and Booking"
- 4 Fix text cut-off on all bottom pages
- 5 Add sharing confirmation toast/popup

● High Impact

Significant UX improvement

- 1 Add doctor verification badge on every summary
- 2 Redesign share page: 'Send To' above content
- 3 Add inline share button per section card
- 4 Add voice input to ai chats and booking
- 5 Make it less text heavy and visually compelling

● Future Iteration

For Fujitsu to implement

- 1 Enrich child profile: photo, DOB, vaccine log and health milestones
- 2 Role-based share presets: full vs. quick summary
- 3 Clarify navigation labels throughout
- 4 Communication presets: Visual / Detailed / Simple

"Oscar and Jesse still have 15+ appointments ahead of them this year. With Pocket Pediatrics inside FHS, they leave each one knowing what happened, what's next, and how to tell their kids.

They no longer feel alone in carrying this cognitive burden and they have time to focus on what truly matters."

The concept reduces cognitive load

Now it's up to Fujitsu to refine the features

14

caregivers

4.0+

every feature, /5

3

additions, built on FHS

Already inside the platform you have. No new system to adopt.

Thank You

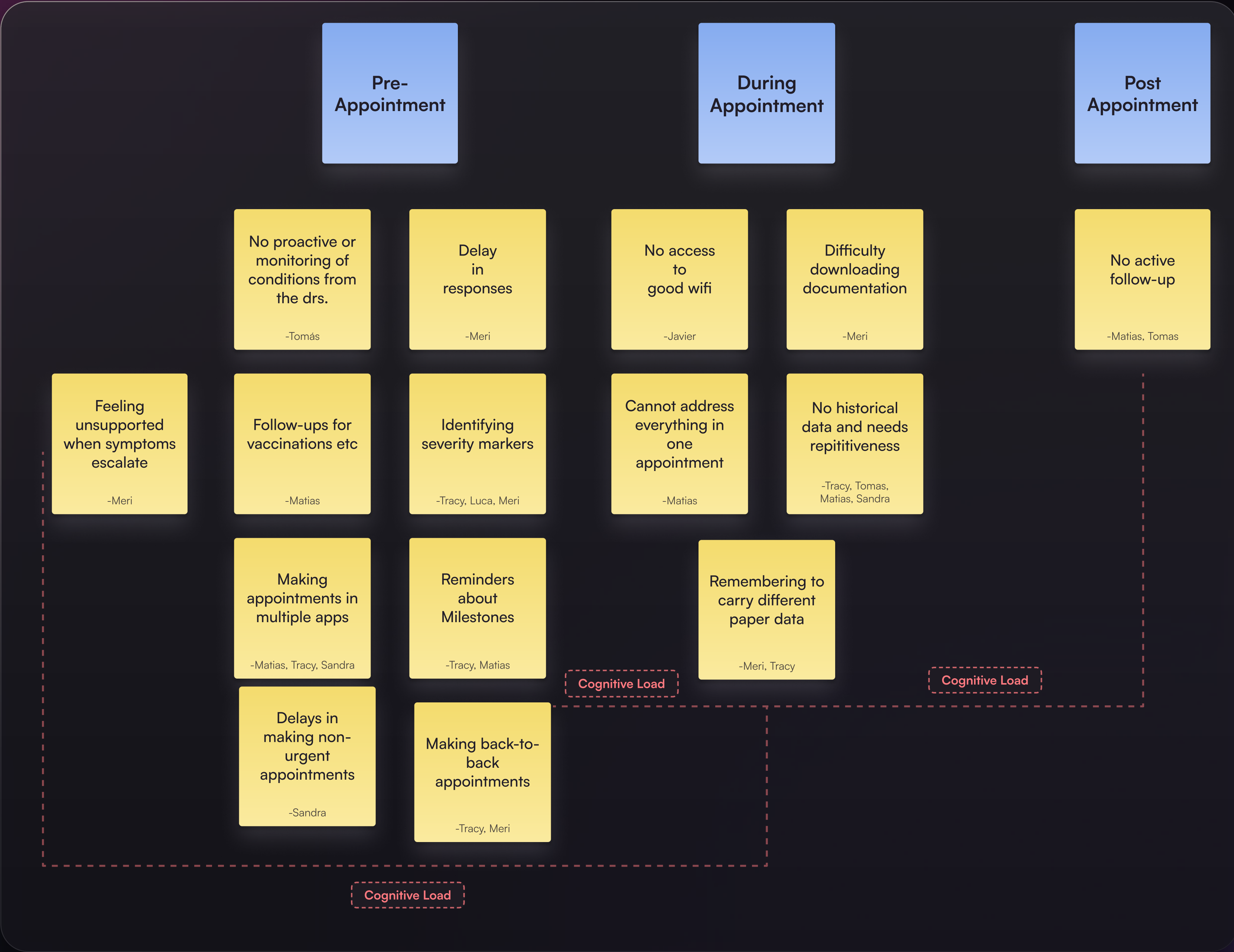
Appendix

Ethnography Map



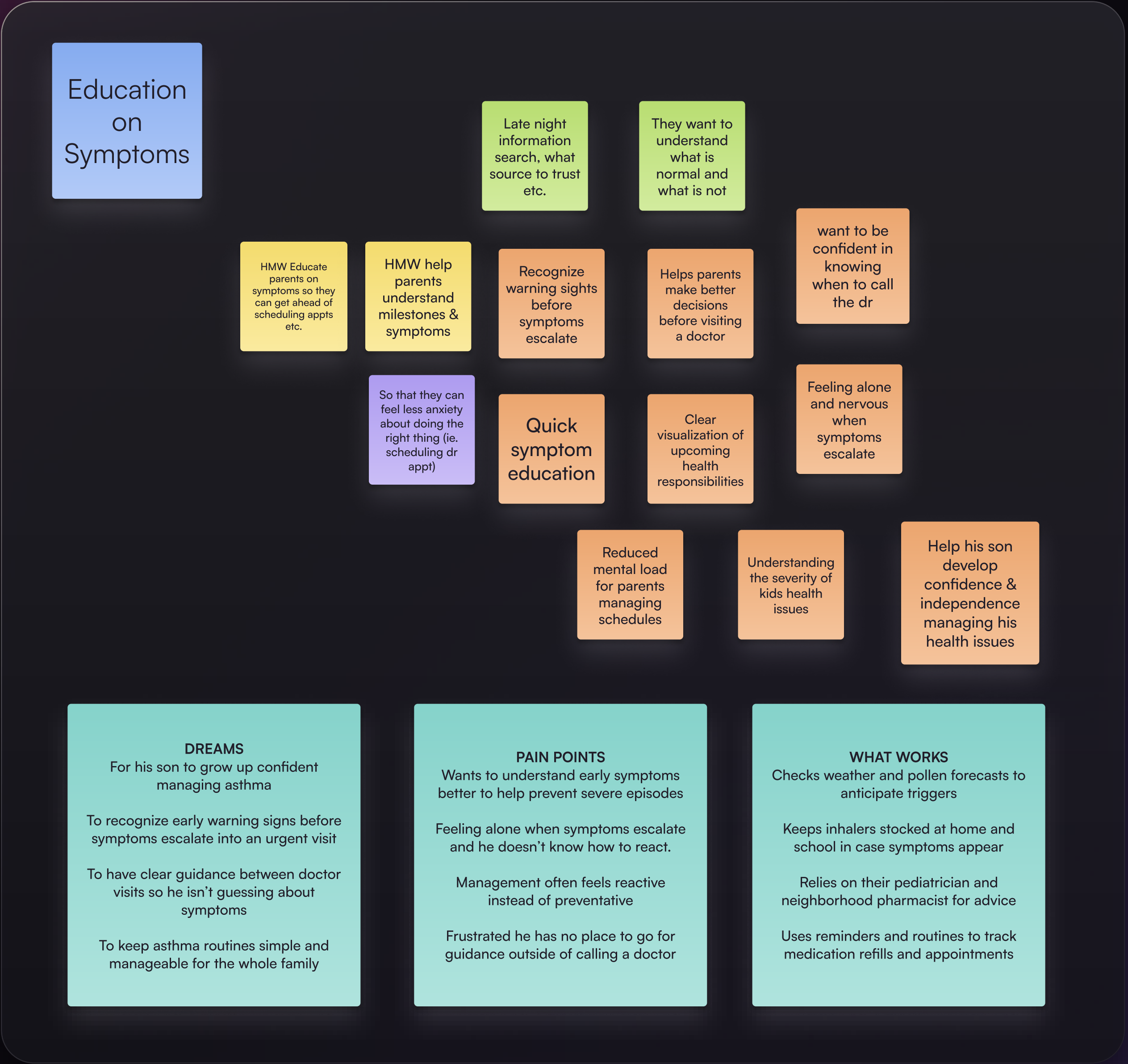
Research Analysis

- Used two strategies to organize data - grouping via post its & organizing by category within an excel sheet
- Noted & highlighted similar issues between interviewees to determine patterns within the pain points.
- Observed most issues discussed occurred in the pre-appointment phase.
- Lots of focus on reminders for milestones like vaccines, when you need to schedule certain appointments etc.
- Concerns about identifying severity of symptoms and illnesses - “is this normal”.
- If things can be addressed pre-appointment then time can be used more strategically during the appointment.



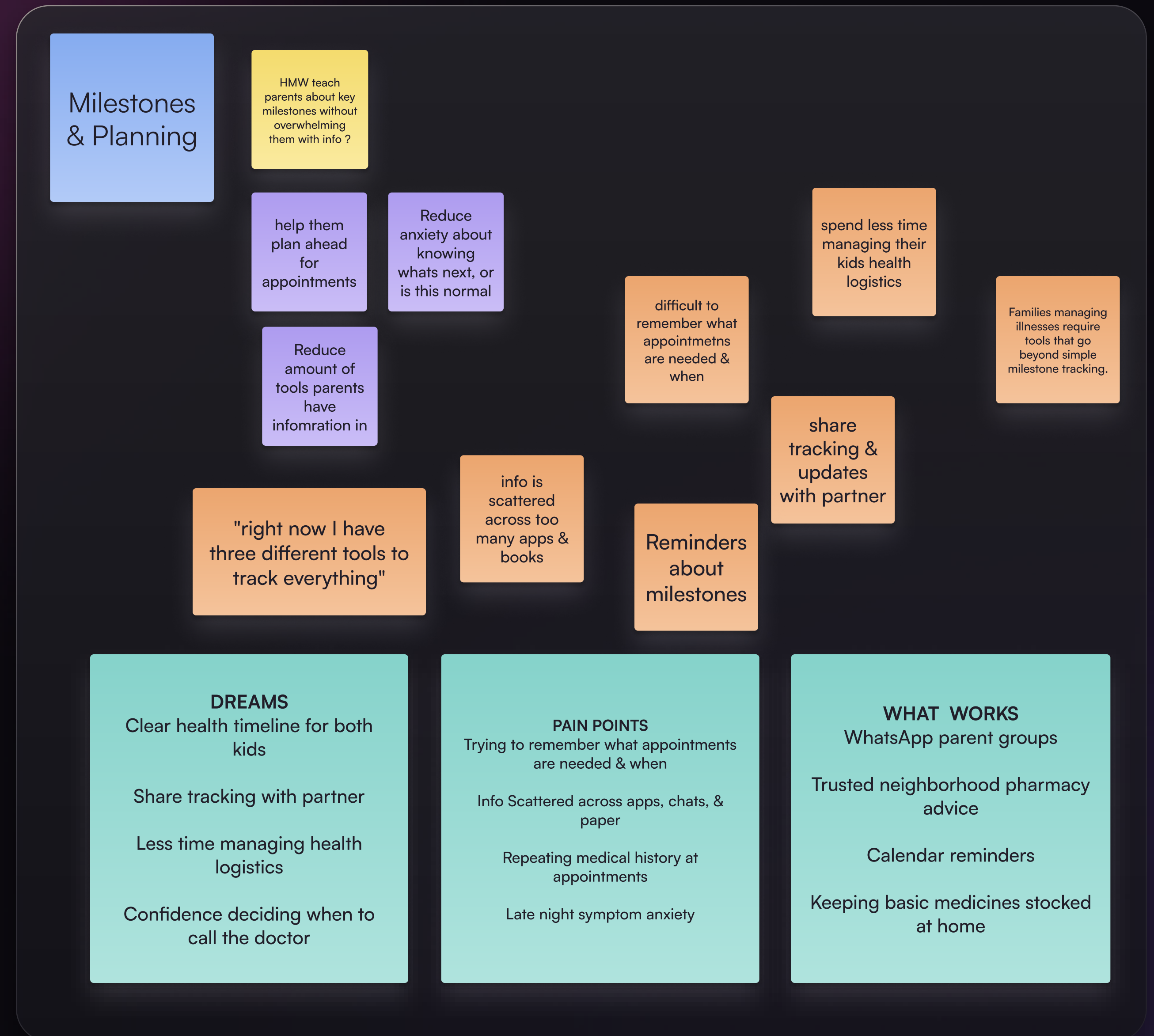
Refining Personas

- This focus is about helping parents **understand symptoms more confidently**.
- Many parents search for answers, especially at night, but are unsure **what information to trust**.
- They need **quick guidance** to recognize warning signs and decide when to call a doctor.
- Right now, symptom management often feels **stressful, lonely, and reactive**.
- So the design opportunity is to provide **trusted, simple, and timely symptom education**.



Refining Personas

- This focus is about how parents manage **milestones and planning** over time.
- Their biggest challenge is that health information and reminders are **fragmented across too many places**.
- Many are already using several tools, but still feel disorganized and anxious.
- They want a clearer system for **what to track, what comes next, and when to act**.
- So the opportunity is to design one experience that supports **planning, reminders, and shared coordination**.



How Might We & Sacrificial Concept Development

Milestones
& Planning

Follow
Up

Education
on
Symptoms

HMW reduce
mental load of
remembering
appointment
follow ups ?

HMW centralize
record keeping of
parents to keep
care giving
organized.

How might we help the
parents and caregivers of
kids 0-10 to create and
understand a creative
place to remember their
milestones?

HMW curate a better
communication
between CG and
doctors of their childs
possible illnesses and
symptoms ?

HMW create an
experience for CG
in order to have
an efficient follow
up for their kids
healthcare ?

How might we reduce
anxiety about the severity
of symptoms through
informal services ?

How might we involve
children in education
of their symptoms to
encourage autonomy
in later years?

How might we help
parents to educate their
children to manage their
own medicines?

How might we organize
milestones so parents
understand what's
necessary vs. flexible?

How might we teach
parents about key
milestones without
overwhelming them with
info ?

How might they create a
solution to improve their
children's milestones and
planning in terms of
healthcare?

HMW create an
efficient medical
past service
which supports
every CG for their
kids ?

How might we help
the parents to
educate their kids to
manage their own
meds ?

Duolingo style
app to teach
people about
symptoms and
+ milestones

DIY Notion
Dashboard

“Weather
dashboard” gives a
“weather outlook”
about milestones
that a child will be
hitting

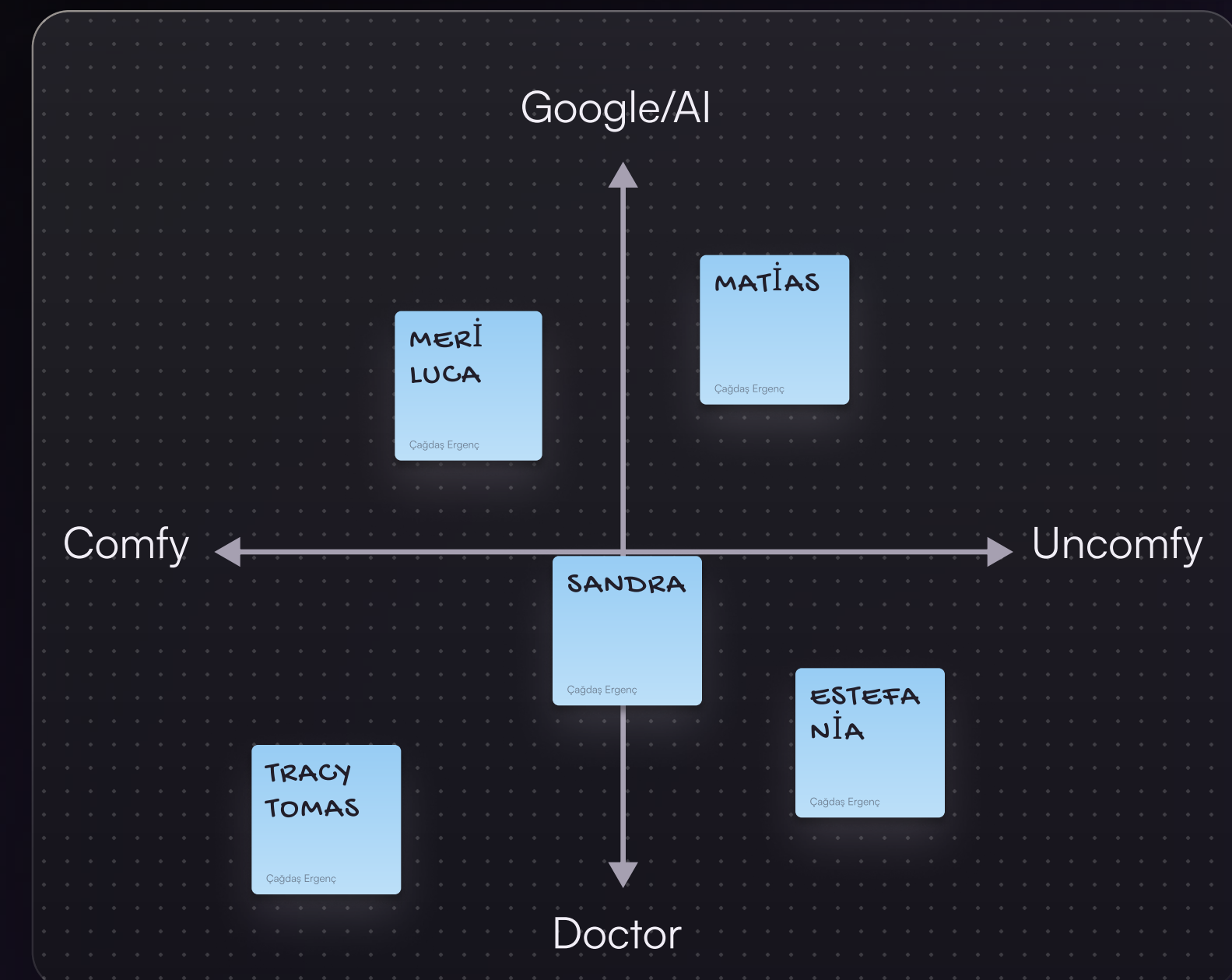
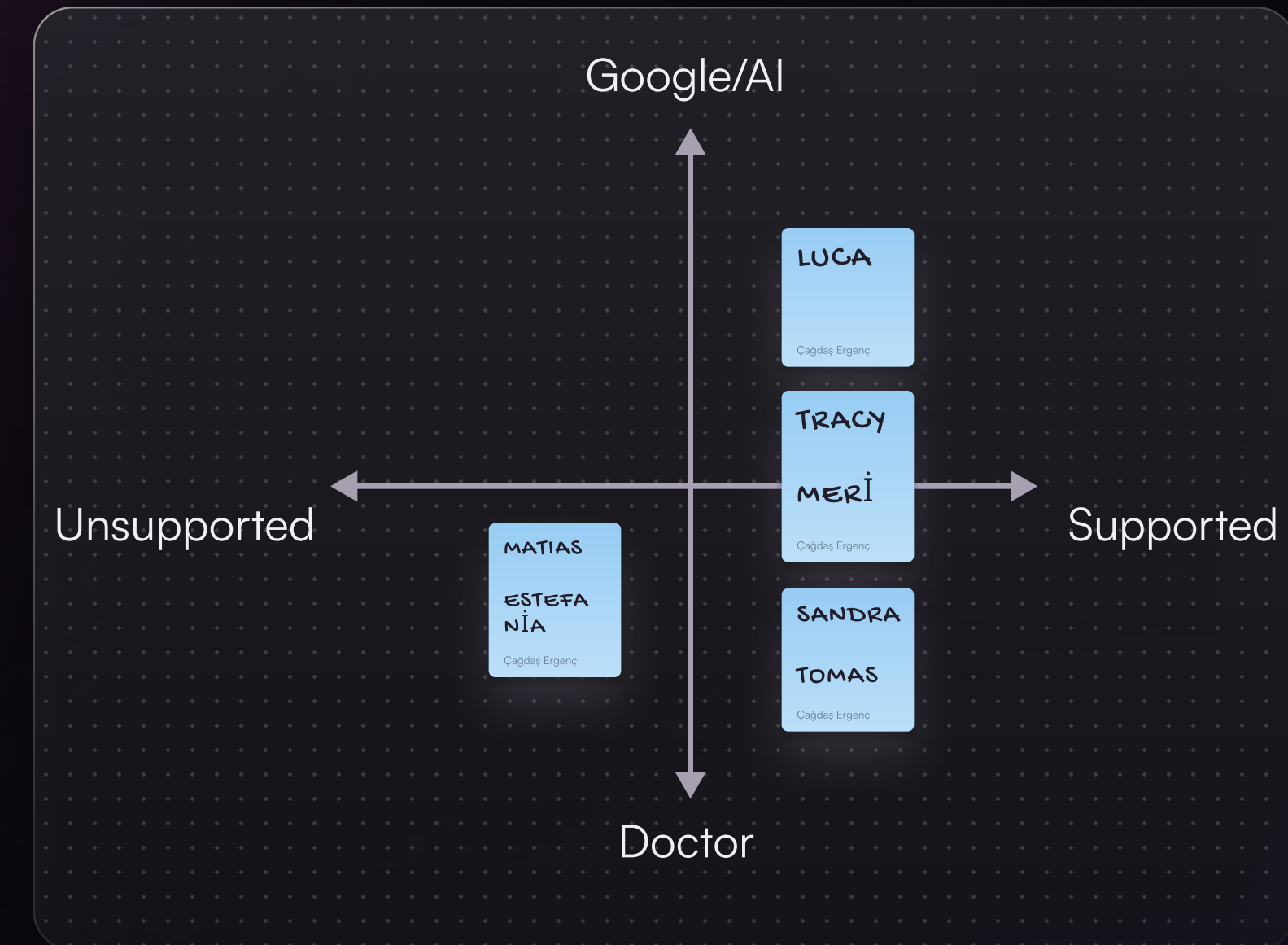
Digital support
groups with 24/7
access - can ask
questions about
symptoms

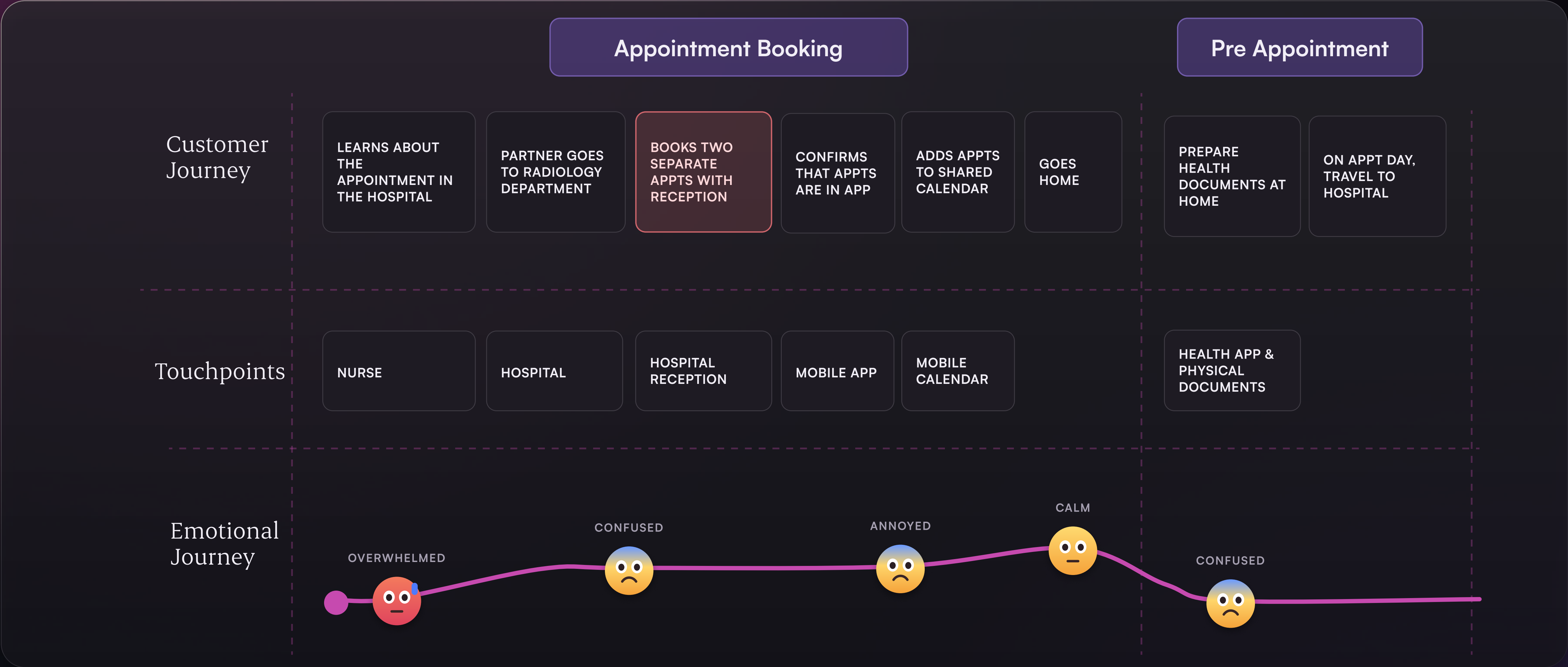
Choose your own
adventure” where
patients can run
through potential
storylines

game where
kids level up
as they hit
milestones

Segmentation

- We segmented participants based on whether they rely more on **doctors or Google/AI** for health guidance.
- We also mapped how **supported or unsupported** and **comfortable or uncomfortable** they feel.
- The results show that most users are not in one extreme group they sit across a **spectrum of trust and comfort**.
- Some participants are more open to **digital or AI support**, while others clearly prefer **doctor-led reassurance**.
- This helps us identify different user needs and shows that future solutions should combine **human trust with accessible digital support**.

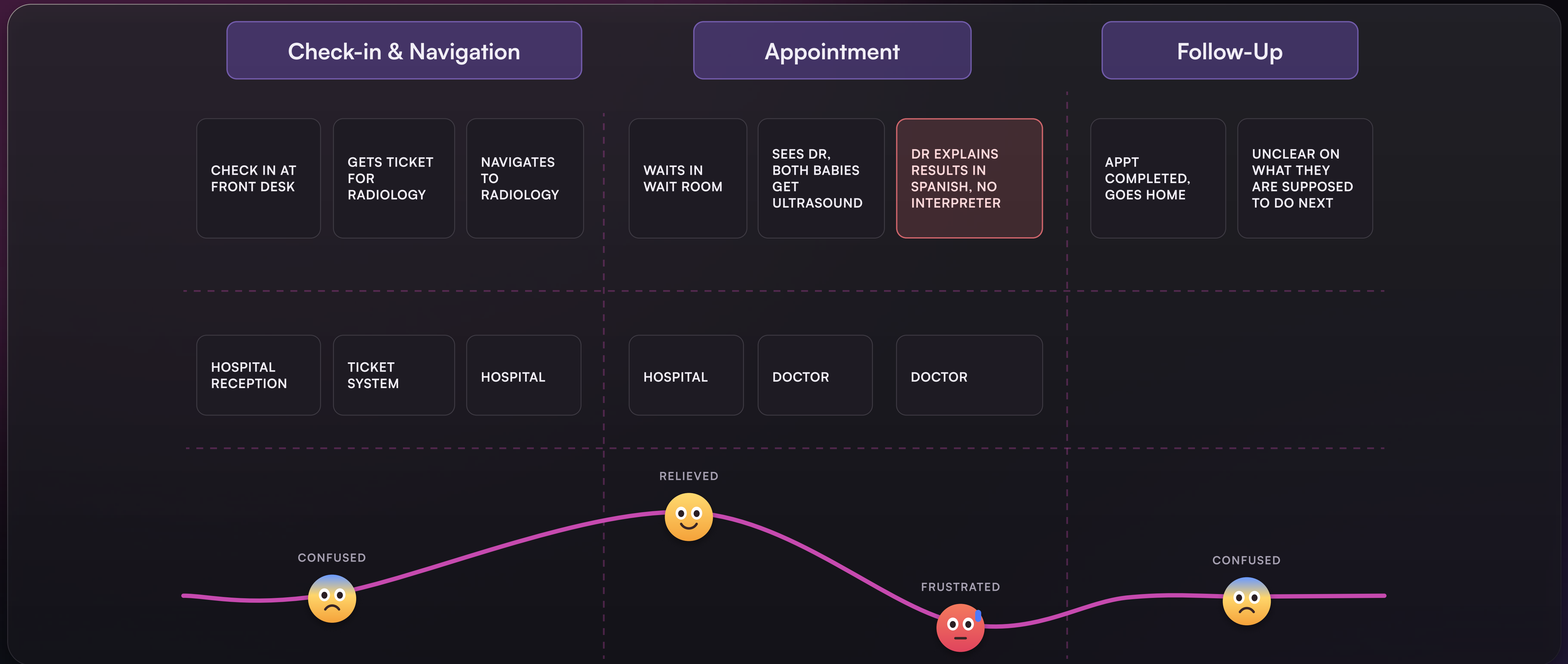




Customer Journey

It is reported that the Spanish public system allocates only **6 minutes** on average for consultations, which leaves limited time for documentation, explanation, and questions.

Sources - [Patients Memory for Medical information](#)



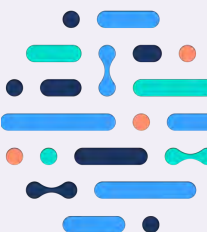
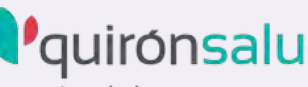
Customer Journey

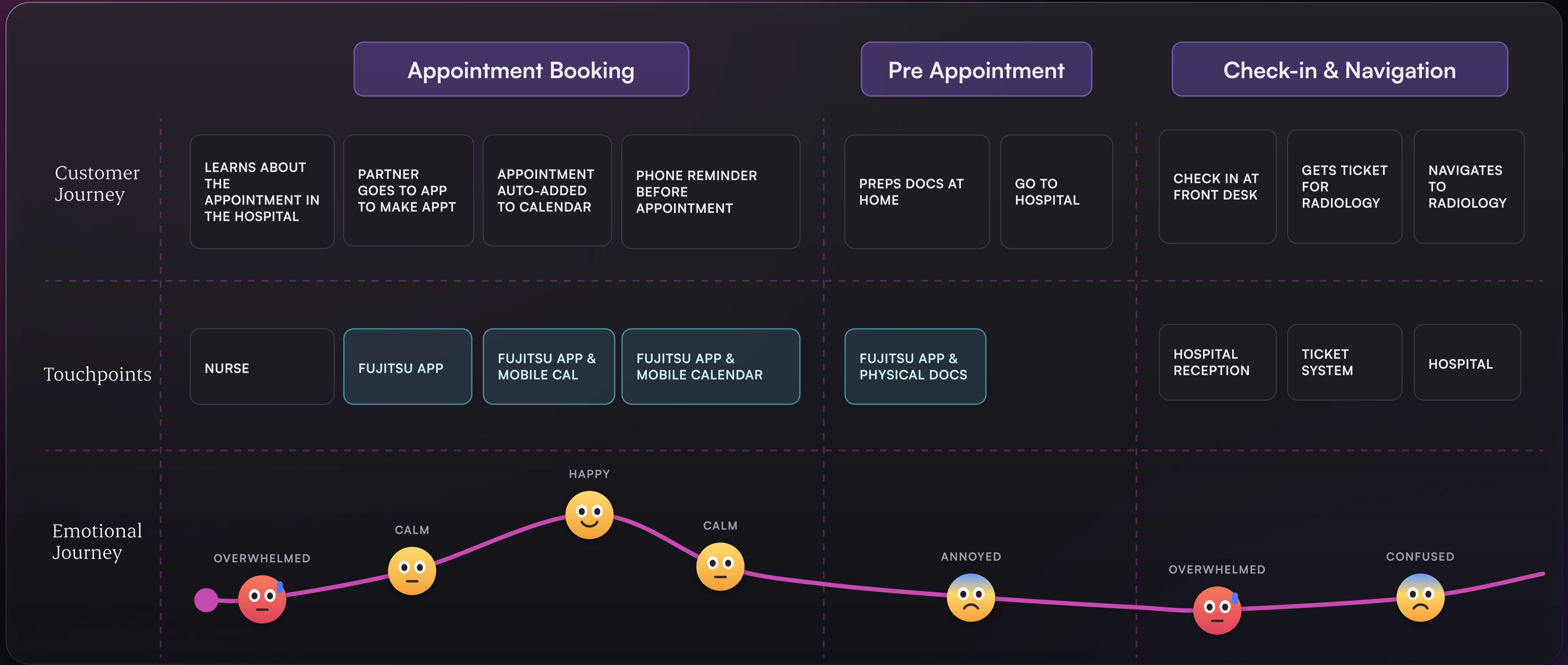
Research shows patients forget **40% to 80%** of medical information immediately after a consultation, and about 50% of what is remembered can be inaccurate.

Sources - [Patients Memory for Medical information](#)

No existing solution delivers all four in one continuous experience

Translation, Scheduling, Follow-up, and AI/Smart UX

Capabilities	 FUJITSU	 ada	 Health Circuit	 La meva /Salut	 KAISER PERMANENTE®	 savia  quirónsalud La salud persona a persona	 Doctoralia
Translation	✓	✓	✗	✗	✗	✗	✗
Scheduling	✓	✗	✗	✗	✗	✗	✓
Follow-up	✓	✗	✓	✗	✓	✗	✗
AI/Smart UX	✓	✓	✓	✗	✓	✗	✗



To be Journey

FHS removes the moments Jesse dreads most — the frantic booking, the language gap, the paper she loses on the way home.

Automated scheduling, real-time translation, and structured follow-up are embedded directly into FHS — so every step is supported, not left to Jesse to figure out alone.

She leaves every appointment confident in the diagnosis and clear on what comes next.

Appointment

WAITS IN
WAIT ROOM

SEES DR,
BOTH BABIES
GET
ULTRASOUND

DR EXPLAINS
RESULTS IN
SPANISH, WITH AI
INTERPRETATION

PATIENT IS SENT A
TRANSCRIPT OF MEETING
& LIST WITH FOLLOW AND
ACTION ITEMS NEEDED

Follow-Up

PATIENT
GOES HOME

APP NOTIFIES
THE USER
WITH THE
DIAGNOSIS

GIVES CLEAR
INSTRUCTIONS
OF NEXT STEPS

WAITING
ROOM

OFFICE

FUJITSU APP & AI
TRANSLATOR

FUJITSU APP & AI
TRANSLATOR

FUJITSU APP

HAPPY

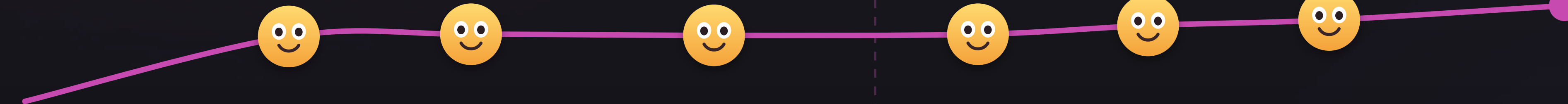
RELIEVED

CONFIDENT

CONFIDENT

SECURE

GUIDED



To be Journey

FHS removes the moments Jesse dreads most — the frantic booking, the language gap, the paper she loses on the way home.

Automated scheduling, real-time translation, and structured follow-up are embedded directly into FHS — so every step is supported, not left to Jesse to figure out alone.

She leaves every appointment confident in the diagnosis and clear on what comes next.

Solution Evaluation

Desirable

Eliminates language barriers with real-time AI translation during appointments.

Automates scheduling, follow-ups, and documentation so patients and doctors can focus on care.

Viable

Adds direct patient value, increasing FHS platform adoption and stickiness.

Reduces administrative burden on doctors, freeing time for higher-value care.

Feasible

FHS's existing AI and interoperability infrastructure become the technical foundation.

US Companies like - Kaiser Permanente already uses AI note-taking and automated patient follow-ups successfully.

How We Tested

14 participants. 2 expert consultations. 5 user journeys. In homes, cafés, and video calls

7

In-Person Interviews

- Face-to-face, think-aloud walkthroughs
- Homes & public places — natural settings
- 30min per session
- Showcased it to people of all ages
- Think-aloud protocol recorded

7

Virtual Run-Throughs

- Working Flutter prototype, used live
- Screen share + verbal commentary
- All sessions recorded
- Enabled broader geographic reach
- Remote via Zoom / Meet

2

Expert Consultations

- Javier — IA & narrative critique
- Mariangela — IA & narrative critique
- Trust & liability concerns surfaced
- High-signal design feedback
- Independent professor reviews

14

Total Participants

- Primary caregivers, grandparents, students
- Ages 24—62, varying digital literacy
- 1 structured + 8 observational sessions
- 2 expert professor reviews

5 CORE JOURNEYS TESTED — MAPPED TO DESIGN PRINCIPLES

T1 Open notification & log in

T2 Find appointment summary

T3 Find follow-up tasks

T4 Share with partner

T5 Create child's story

Executive Summary

01

Overall Experience

3.8 / 5

Avg across clarity, completeness & trust scored by 14 participants

02

Top Positives

- Sharing rated #1 tool — fast, intuitive, useful
- Story feature consistently loved by all participants
- Content completeness strong — information thorough
- Structured Notes clearer than full Summary

03

Top Negatives

- No sharing feature directly from Summary and Notes page
- Summary vs. Notes distinction unclear to most users
- No sharing confirmation after sending
- Schedule + Appointments tabs overlap in purpose

04

Areas of Attention

- Text size & readability on smaller screens
- AI trust layer missing — need confirmation of doctor review
- Graphs, tools & tips
- Order of appointment and schedule dates

Appointment Summary & Structured Notes

4.6 /5
Summary Clarity

4.6 /5
Info Completeness

✓ WHAT'S WORKING

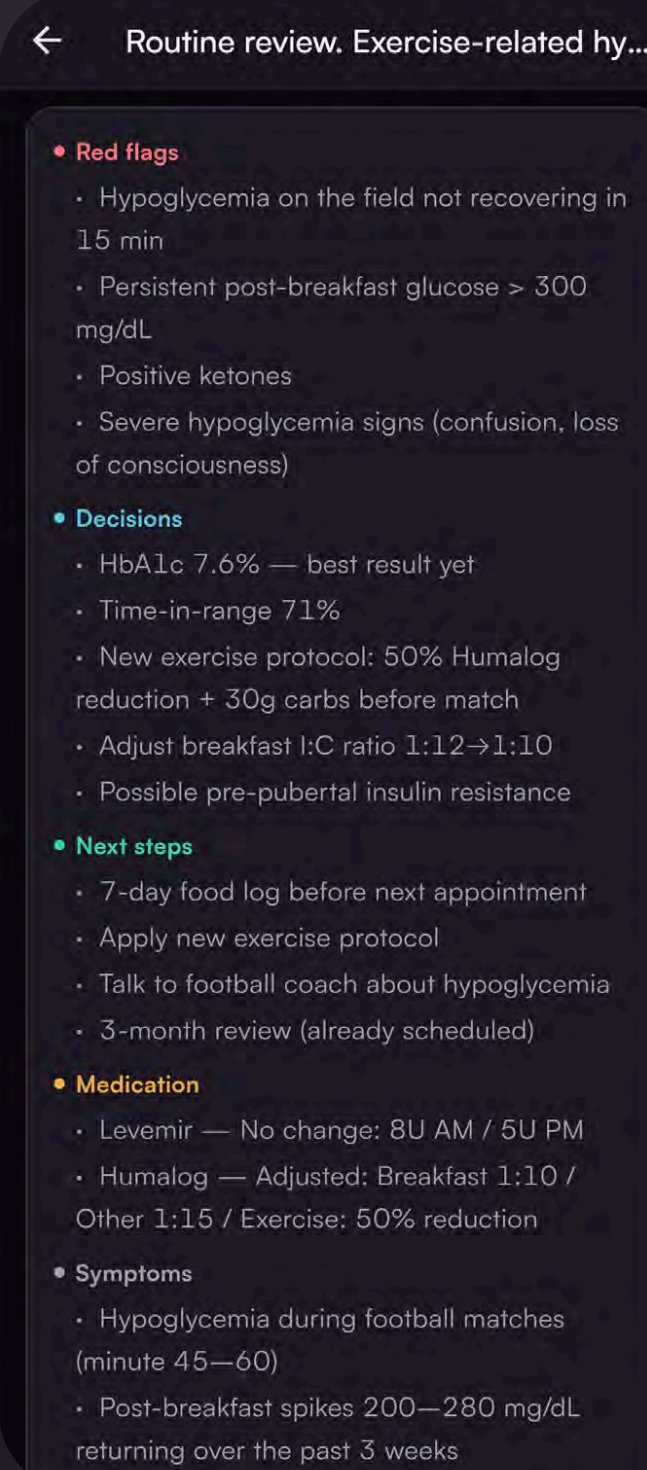
- ✓ Structured Notes felt very clear to users
- ✓ Content completeness strong — information thorough
- ✓ Users could locate visit-specific information quickly
- ✓ Overall satisfaction above average across both metrics

✗ WHAT NEEDS FIXING

- ✗ Summary felt overwhelming — too much info visible at first glance
- ✗ Medical jargon in summary hard to understand
- ✗ No tap-to-expand for plain-language explanations
- ✗ Confused between 'recent history' and 'follow-up' sections

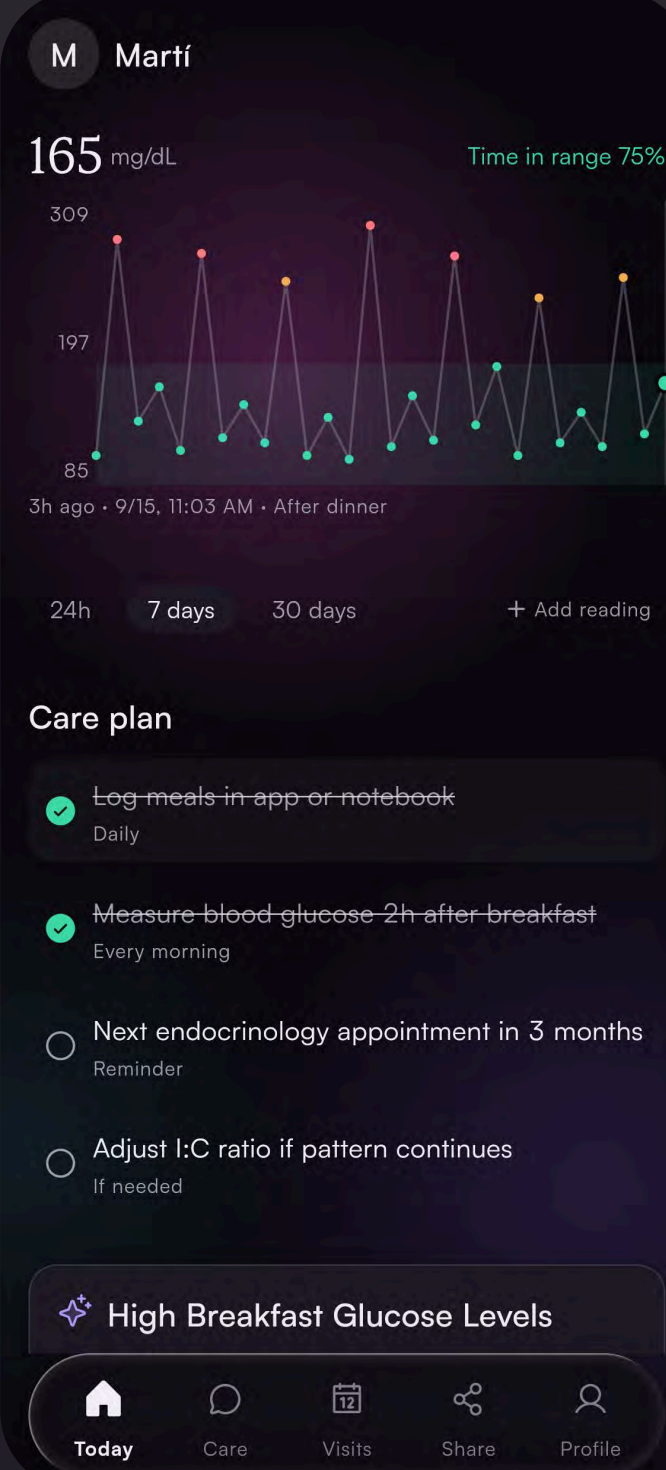
→ RECOMMENDATIONS

- Flip hierarchy: Structured Notes first, Summary collapsed
- New order: Red Flags → Next Steps → Medications → Notes
- Add tap-to-expand glossary on every medical term



Structured Notes

High level notes of last appointment



Care Plan

Care plan checklist for caregiver and doctor tracking

Sharing Feature

4.8 /5
#1 Rated Feature

✓ WHAT'S WORKING

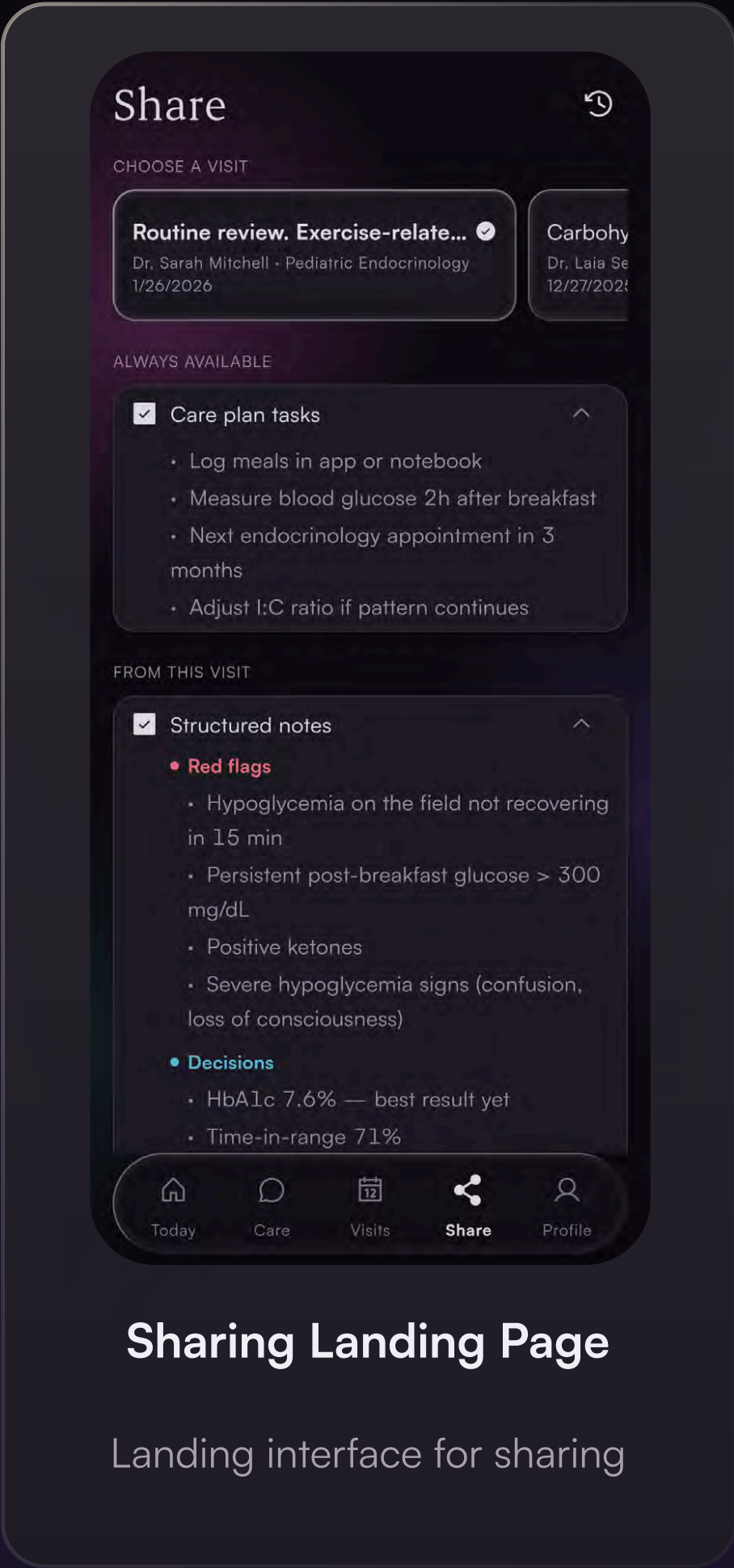
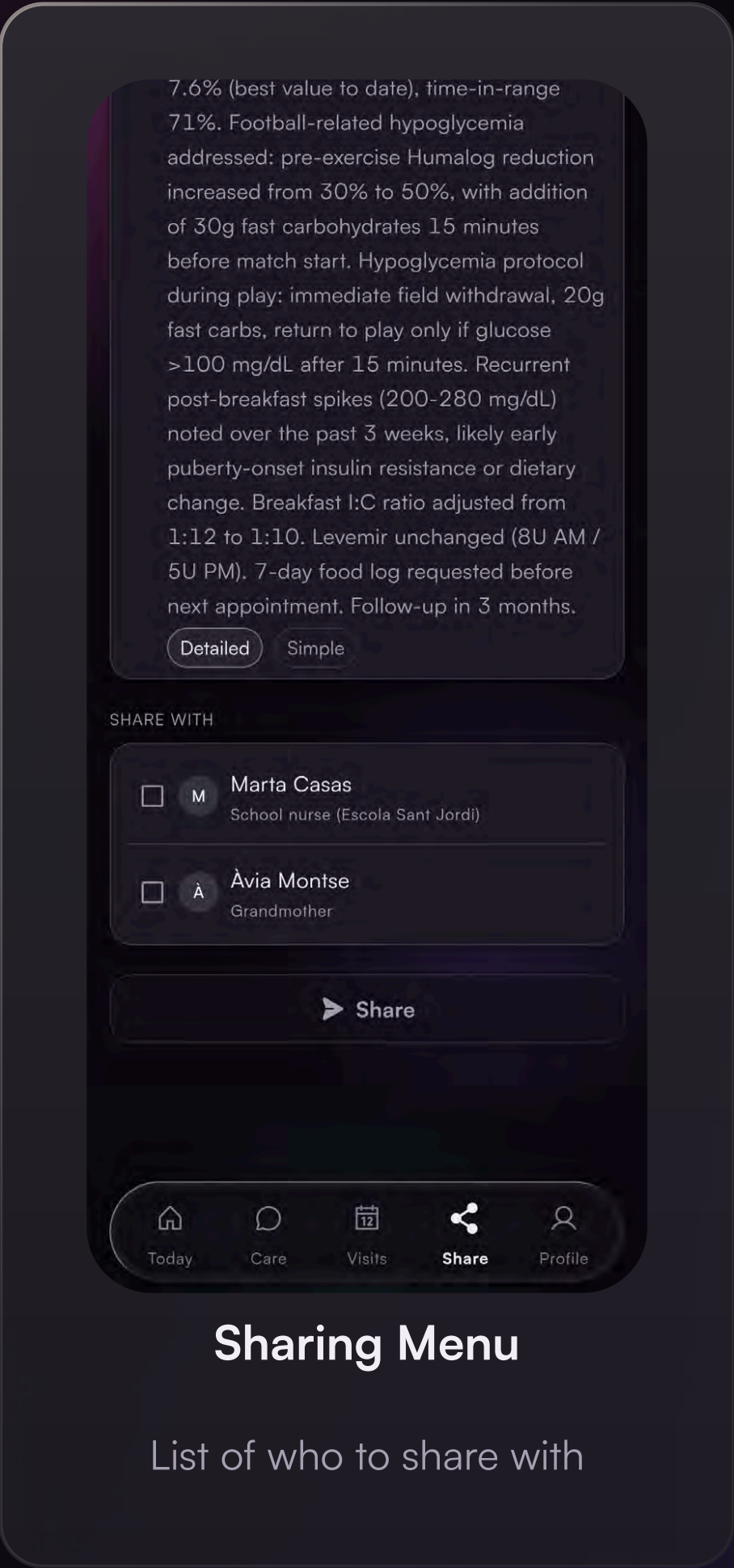
- ✓ Highest rated feature — all participants scored 4 or 5 out of 5
- ✓ Intuitive and fast — caregivers found core sharing flow immediately
- ✓ Care circle concept resonated with interviewees
- ✓ Sharing tab structure clear enough that most users completed without friction

✗ WHAT NEEDS FIXING

- ✗ Share button on separate tab — 7/14 users expected on summary screen
- ✗ No confirmation after sharing — users couldn't tell if info was actually sent
- ✗ "Send To" is hidden below the content being shared
- ✗ Share page sends everything by default
- ✗ No role presets — need full report for partner, quick summary for grandparent

→ RECOMMENDATIONS

- Flip hierarchy: Structured Notes first, Summary collapsed
- New order: Red Flags → Next Steps → Medications → Notes
- Add tap-to-expand glossary on every medical term



The Story Feature

4.0 /5
Story Feature Avg

✓ WHAT'S WORKING

- ✓ Consistently loved across all participants
- ✓ Caregivers agree it will help explain medical events to children
- ✓ Simple and precise — not overwhelming
- ✓ Tone felt warm and age-appropriate for kids

✗ WHAT NEEDS FIXING

- ✗ Wrong location — should be on follow-up screen, not current placement
- ✗ Covers today's appointment only — not full illness history
- ✗ Reads as a passive story, not an active parent script
- ✗ Name "Simple Story" doesn't signal the feature's purpose clearly

→ RECOMMENDATIONS

- Rename "Simple Story" → "Talk to [Child's Name]"
- Move feature to the follow-up screen
- Reframe copy as parent-facing script, not passive narrative

